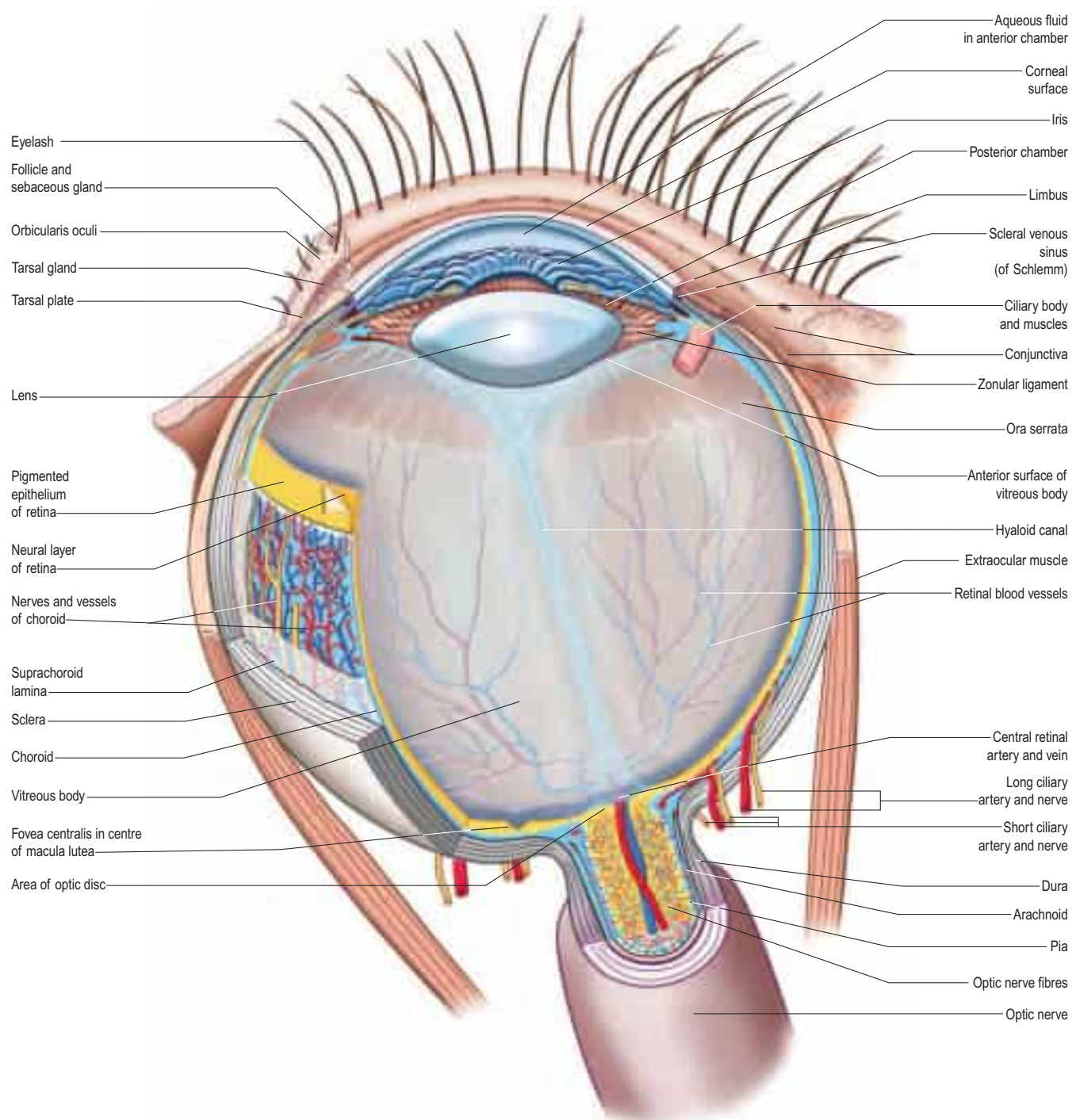


The outer surface of the eye is composed of parts of two spheres with different radii. The anterior segment, part of the smaller sphere, is formed by the transparent cornea and accounts for approximately 7% of the ocular surface. The posterior segment of the eyeball is part of the larger sphere formed by the opaque sclera (Fig. 42.1).

Internal to the sclera is a vascular, pigmented layer consisting of three continuous parts that collectively make up the uveal tract: a thin

choroid lying posteriorly, a thicker ciliary body and an anterior iris that is displaced from the outer coat and terminates at the pupillary aperture. The internal surface of the choroid is covered by the photo-sensory retina, which terminates anteriorly at the ora serrata; the latter also marks the junction between the ciliary body and choroid. The vasculature of the choroid supplies nutrients to the avascular outer retina.



The ocular lens is located immediately behind the iris and is suspended from the ciliary body via zonular fibres (see Fig. 42.1). Smooth muscles within the ciliary body regulate the tension exerted on the elastic lens and hence determine its shape, thereby adjusting the focus of the eye in the process of accommodation. The iris, which does not allow the transmission of light due to a heavily pigmented posterior surface, also contains smooth muscle, allowing it to regulate the size of the pupillary aperture.

The iris and lens separate the eye into three chambers. The largest, the vitreous chamber, is filled with the gel-like vitreous humour, and lies posterior to the lens, comprising about two-thirds of the volume of the eye (see Fig. 42.1). The spaces between the lens and iris, and the iris and cornea, are, respectively, the posterior and anterior chambers of the eye. Both are filled with aqueous humour, which is produced by the epithelium of the ciliary body, travels through the pupil, circulates in the anterior chamber and is drained principally through the canal of Schlemm at the iridocorneal filtration angle. Aqueous humour provides metabolic support to the avascular lens and cornea.

The sole purpose of the eyes and their associated structures within the bony orbit is to form a good image on a healthy retina. The photoreceptors of the retina transduce the optical radiation into neurobiological activity and other cells within the retina then begin to process the output of the photoreceptors. The retinal signal is transferred via the optic nerve, along the visual pathway (see Fig. 42.32) to various regions within the brain, where further processing results in visual perception.

OUTER COAT

The fibrous outer coat of the eye consists of the opaque posterior sclera and the transparent anterior cornea. Together they form a semi-elastic protective capsule enclosing the eye, which, when made turgid by intraocular pressure, determines the optical geometry of the eye and ensures its shape is not distorted when it moves. The sclera also provides attachments for the extraocular muscles, and its smooth external surface rotates easily on the adjacent tissues of the orbit when these muscles contract (Ch. 41). The opacity of the sclera helps ensure that only light that enters the eye through the pupil reaches the retina. The cornea, on the other hand, not only admits light but also its covering tear film is the major refractive surface of the eye.

SCLERA

The sclera accounts for approximately 93% of the outer coat of the eye. Anteriorly, it is continuous with the cornea at the corneoscleral junction (see Fig. 42.1). It is punctured by a number of foramina containing nerves and blood vessels, most notably the optic foramen, which lies 3 mm medial to the midline and 1 mm below the horizontal, and houses the optic nerve. Smaller openings contain anterior ciliary arteries that penetrate anteriorly, vortex veins that cross the sclera equatorially, and the long and short ciliary nerves and arteries that enter posteriorly. There is considerable individual variation in scleral dimensions. The sclera is thickest at the posterior pole (approximately 1 mm) and decreases anteriorly, reaching a minimum equatorially at about half this thickness. It also thins approaching the optic nerve. The sclera is thinner when the eye is elongated in myopia.

The external surface of the sclera is covered by a delicate episcleral lamina of loose fibrovascular tissue, which contains sparse blood vessels and is in contact with the inner surface of the fascial sheath of the eyeball. Anteriorly, the external scleral surface is covered by conjunctiva, which is reflected on to it from the posterior surfaces of the eyelids. The scleral internal surface adjacent to the choroid is attached to it by a delicate fibrous layer, the suprachoroid lamina, which contains numerous fibroblasts and melanocytes. Anteriorly, the inner sclera is attached to the ciliary body by the lamina supraciliaris. Posteriorly, the sclera is pierced by the optic nerve. Here, the outer half of the sclera turns back to become continuous with the dura mater, while the inner half is modified to form a perforated plate, the lamina cribrosa sclerae. The optic nerve fascicles pass through these minute orifices, while the central retinal artery and vein pass through a larger, central aperture. The lamina cribrosa sclerae is the weakest part of the sclera and bulges outwards (a cupped disc) when intraocular pressure is raised chronically, as in glaucoma.

Like the cornea, the scleral stroma is composed mainly of densely packed collagen embedded in a matrix of proteoglycans, which are mixed with occasional elastic fibres and fibroblasts. However, in contrast to the cornea, scleral collagen fibrils show a large variation in diameter and spacing, and the lamellae branch and interlace extensively. This arrangement of fibres results in increased light scatter, which is responsible for the opaque, dull-white appearance of the sclera, and also imparts a high tensile strength to the sclera to resist the pull of the extraocular muscles and contain the intraocular pressure. Collagen fibre bundles are arranged circumferentially around the optic disc and the orifices of the lamina cribrosa. The fibres of the tendons of the recti intersect scleral fibres at right angles at their attachments, and then interlace deeper in the sclera. Collagen fibres of the scleral spur are orientated in a circular fashion, and there is an increased incidence of elastic fibres here (Figs 42.2–42.3A). Although the sclera acts as a conduit for blood vessels, scleral vessels are few and mainly disposed in the episcleral lamina, especially close to the limbus. Its nerve supply is surprisingly rich, accounting for the intense pain associated with scleral inflammation (Watson and Young 2004).

Scleral development is under active regulation to ensure an eye of the correct axial length to produce a focused image (Wallman and Winawer 2004).

Filtration angle and aqueous drainage

Aqueous humour is produced by the ciliary epithelium; it passes through the pupil and circulates within the anterior chamber, supplying the avascular cornea and lens with nutrients and removing metabolic waste products. It drains from the eye mainly through the trabecular meshwork into the canal of Schlemm at the iridocorneal (filtration) angle (see Figs 42.2–42.3), formed between the posterior aspect of the corneoscleral limbus and the periphery of the iris. Differences in drainage angle morphology most probably underlie ethnic variations in the prevalence of primary angle closure glaucoma (Wang et al 2012).

The trabecular meshwork is continuous anteriorly with Descemet's membrane and the endothelium of the cornea. Aqueous humour filters from the anterior chamber through interconnected spaces among loose trabecular fibres, most of which are attached to the anterior, external aspect of the scleral spur. Of the remainder, most are continuous with longitudinal fibres of the ciliary muscle, some of which attach to the

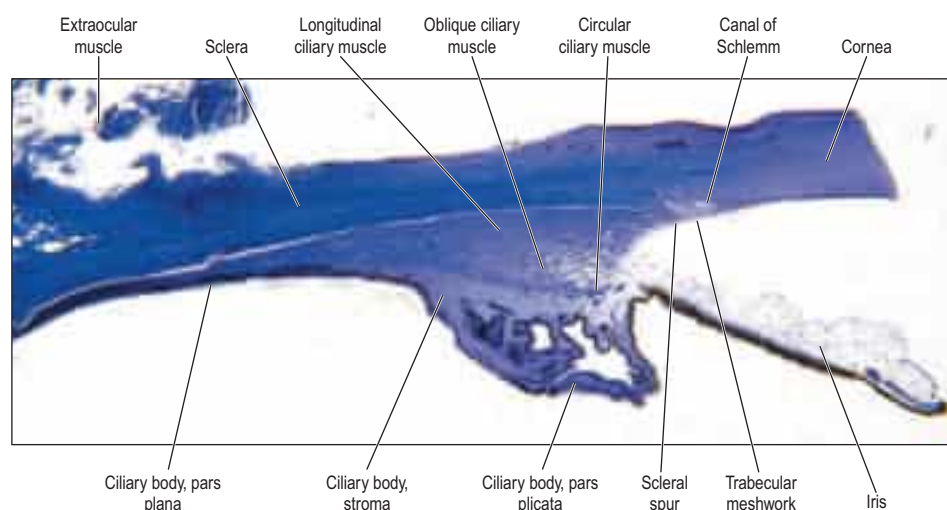


Fig. 42.2 A meridional section through the anterior eye, highlighting the ciliary body and drainage apparatus.

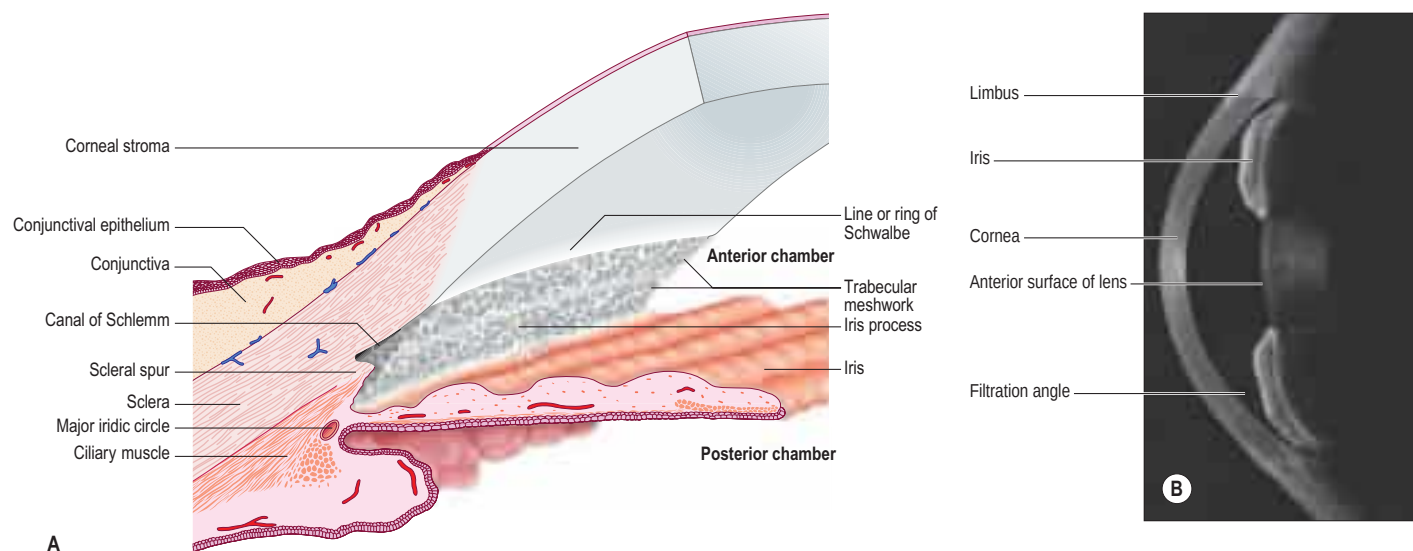


Fig. 42.3 **A**, A schematic representation of the tissues bordering the anterior chamber angle. The trabecular meshwork is an anulus of tissue spanning the angle; its meshes are shown in transverse section opposite the canal of Schlemm and partly attached to the scleral spur. A single iris process bridges the angle, connecting the trabecular meshwork to the anterior tissues of the iris. **B**, An optical coherence tomography (OCT) *in vivo* image of the anterior segment of a 60-year-old Caucasian female taken in dim room light. (B, Courtesy of Mr Gus Gazzard, Consultant Ophthalmic Surgeon, Moorfields Eye Hospital, London.)

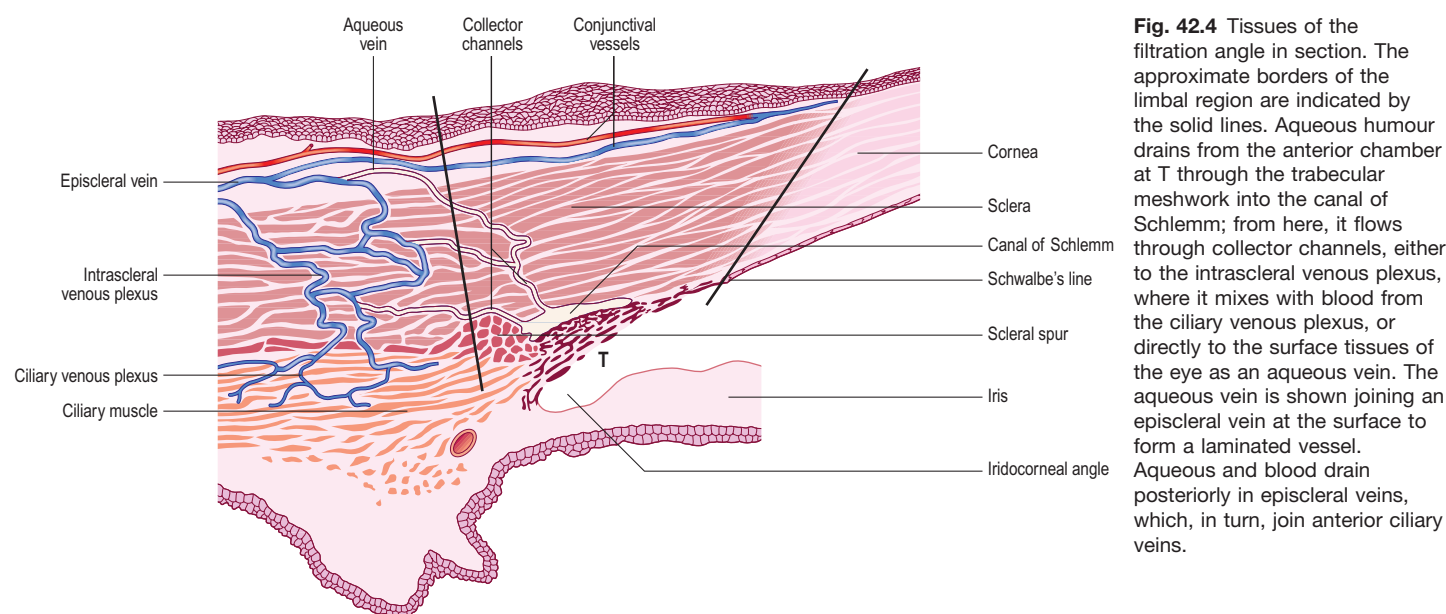


Fig. 42.4 Tissues of the filtration angle in section. The approximate borders of the limbal region are indicated by the solid lines. Aqueous humour drains from the anterior chamber at T through the trabecular meshwork into the canal of Schlemm; from here, it flows through collector channels, either to the intrascleral venous plexus, where it mixes with blood from the ciliary venous plexus, or directly to the surface tissues of the eye as an aqueous vein. The aqueous vein is shown joining an episcleral vein at the surface to form a laminated vessel. Aqueous and blood drain posteriorly in episcleral veins, which, in turn, join anterior ciliary veins.

posterior internal aspect of the scleral spur. The trabecular meshwork provides sufficient resistance to aqueous humour outflow to generate an intraocular pressure of approximately 15 mmHg. It also acts as a filter and has the capacity to phagocytose particulate matter, although overloading may contribute to the pathogenesis of various forms of obstructive secondary glaucoma (p. 697).

The canal of Schlemm (sinus venosus sclerae) is an annular endothelial canal located near the internal surface of the sclera close to the limbus. In section, the canal appears as an oval cleft, with an outer wall that grooves the sclera. Posteriorly, the cleft extends as far as a rim of scleral tissue, the scleral spur, which, in section, forms a triangle with its apex directed forwards (see Figs 42.2–42.3A). The canal of Schlemm may be double or multiple in part of its course, and its walls are constructed of a continuous single thin endothelial layer. Passage of aqueous humour to the canal probably occurs via giant pinocytotic vacuoles and associated transcellular pores, which form on the inner face of the endothelium and discharge into the canal at the outer face. Aqueous humour passes through a plexus of fine intrascleral vessels that connect the canal of Schlemm with episcleral veins (Fig. 42.4). Normally, the canal does not contain blood; pressure gradients prevent the reflux of blood, even though the channels between the canal and veins have no valves. However, in venous congestion, blood may enter the canal; the continuous endothelial outer wall of the canal prevents further reflux.

An alternative route for aqueous outflow, the uveoscleral pathway, has been described (Fautsch and Johnson 2006, Goel et al 2010). Estimates of aqueous drained by this route in humans vary from under 5%

to as much as 54% (Alm and Nilsson 2009), and decrease with age. Since there is no epithelial barrier between the anterior chamber and the ciliary body, aqueous humour is able to enter the loose connective tissue in front of the ciliary muscle and pass between the muscle fibres into the supraciliary and suprachoroidal spaces, where, potentially, it can be absorbed by vessels that drain the uvea. Clinically, aqueous outflow via this route can be increased by topical prostaglandin analogues, which are first-line therapy for the treatment of open angle glaucoma.

CORNEA

The avascular cornea is the anterior transparent part of the outer coat. Convex anteriorly, it projects from the sclera as a dome-shaped elevation with an area of 1.1 cm², forming approximately 7% of the external tunic area. Since it is more curved (average radius, $r=7.8$ mm) than the sclera ($r=11.5$ mm), a slight sulcus sclerae marks the corneoscleral junction (limbus). The high corneal curvature of the neonate cornea decreases to reach adult levels within a relatively short space of time (Friling et al 2004).

The cornea is approximately 670 μ m thick close to the corneoscleral junction, and 520 μ m at its centre. At the nasal and temporal limbus, the transition from cornea to sclera occurs in a line that is approximately perpendicular to the cornea; this transition occurs more obliquely superiorly and inferiorly, with the sclera overlapping the cornea to a greater extent anteriorly. Consequently, when viewed from

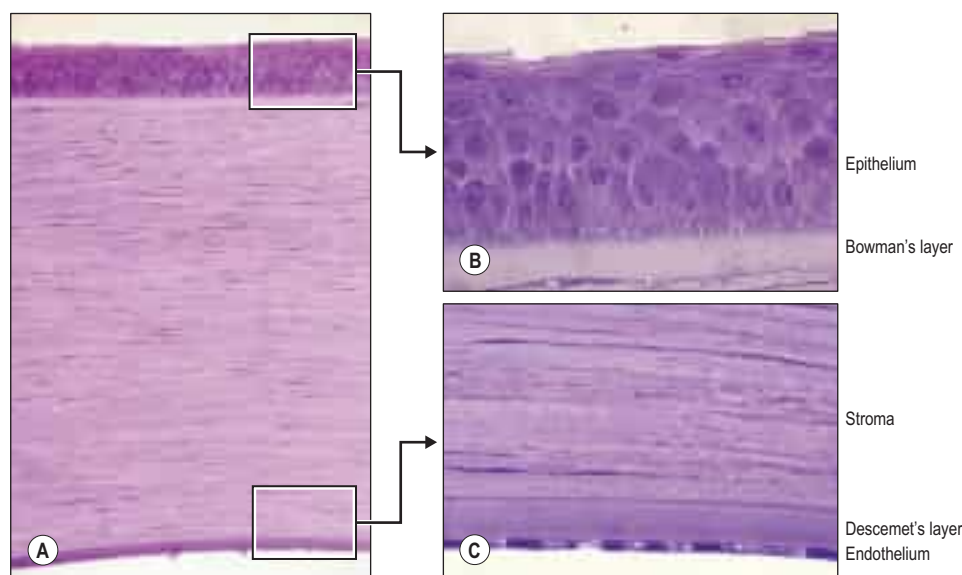


Fig. 42.5 **A**, A low-power light micrograph of the five layers of the cornea. **B**, A high-power light micrograph of the corneal epithelium and Bowman's layer. **C**, A high-power light micrograph of the corneal endothelium, Descemet's layer and part of the stroma. The darkly staining cells in the stroma are keratocytes.

in front, the corneal perimeter is slightly elliptical, as its horizontal diameter (11.7 mm) is a little greater than its vertical (10.6 mm). Its posterior perimeter is circular (diameter 11.7 mm).

Since the cornea carries most of the optical power of the eye, changes to its shape can have a significant impact on the refractive state of the eye (p. 699). This has been used to advantage in a number of techniques for correcting refractive error such as laser refractive surgery.

Microscopically, the cornea consists of five layers: corneal epithelium, anterior limiting lamina (Bowman's layer), substantia propria (stroma), posterior limiting lamina (Descemet's membrane) and endothelium, arranged anteroposteriorly (Fig. 42.5).

Corneal epithelium

The corneal epithelium accounts for approximately 10% of the corneal thickness (50 μm). It usually consists of 5–6 layers of cells (Fig. 42.5B) that protect the ocular surface from mechanical abrasion, form a permeability barrier (to small molecules, water and ions) and prevent the entry of pathogens. The deepest cells are columnar with flat bases, rounded apices and large round or oval nuclei. Outside these are 2–3 layers of polyhedral (often wing-shaped) cells. In the more superficial layers, the cells become progressively flatter and present a smooth, optically perfect surface. Scanning electron microscopy of surface cells reveals extensive finger-like and ridge-like projections (microvilli and microplicae). A complex network of tight junctions links the superficial cells, consistent with their barrier function.

Anterior limiting lamina (Bowman's layer)

The anterior limiting lamina lies behind the corneal epithelium. It contains a dense mass of collagen fibrils set in a matrix similar to that of the substantia propria. The lamina is 12 μm thick, is readily distinguishable from the substantia propria because it does not contain fibroblasts, and appears amorphous by light microscopy (see Fig. 42.5B).

Substantia propria (stroma)

The substantia propria is approximately 500 μm thick and forms the bulk of the cornea. It is a compact and transparent layer, composed of 200–250 sequential lamellae, each made up of fine parallel collagen fibrils mainly of type I collagen. Flat dendritic interconnecting fibroblasts (keratocytes) form a coarse mesh between the lamellae. Alternate lamellae are typically orientated at large angles to each other (Fig. 42.6). X-ray diffraction studies indicate that they run along two preferred directions – superior–inferior and nasal–temporal – to account for the additional tensile stress exerted by the recti along these meridians (Boote et al 2005). Each lamella is approximately 2 μm thick and of variable width (10–250 μm or, rarely, more). In the central cornea, fibrils within the lamellae have a similar diameter of approximately 31 nm. This increases slightly with age and approaching the limbus. The small size of the fibrils (much smaller than the wavelength of light), along with the regularity of their spacing (maintained by collagen–proteoglycan interactions) and the careful control of corneal hydration,

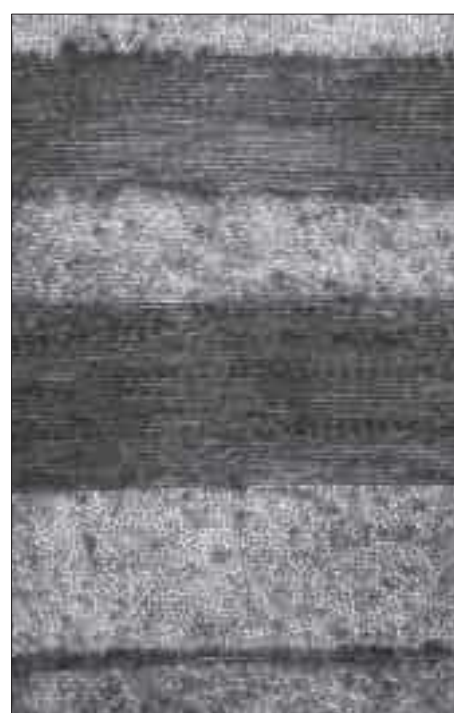


Fig. 42.6 An electron micrograph of the stroma of the human cornea, showing the geometric precision of the alternation in direction of adjacent layers of fibres. (Courtesy of John Marshall, St Thomas' Hospital, London.)

are the principal factors that determine stromal transparency. Light scattered by the collagen fibrils is eliminated by destructive interference in all directions other than forwards.

Posterior limiting lamina (Descemet's layer)

The posterior limiting lamina covers the substantia propria posteriorly. It is thin and apparently homogeneous, and is regarded as the basement membrane of the endothelium (see Fig. 42.5C). Some 4 μm thick at birth, it may increase to 12 μm by the eighth decade.

Endothelium

The endothelium covers the posterior surface of the cornea and consists of a single layer of squamous cells (see Fig. 42.5C) with prominent interdigitations between adjacent cells, which are also connected by tight and gap junctions. When viewed *en face*, the endothelium appears as a mosaic of polygonal (typically hexagonal) cells. As these cells have a limited capacity for mitosis, in response to pathology, trauma, age and prolonged contact lens wear, the endothelial mosaic becomes less regular, and shows a greater variation in cell size (pleomorphism) and shape (polymegathism), as cells spread to fill gaps caused by cell loss. The numerous mitochondria and prominent rough endoplasmic reticulum within these cells reflects their high metabolic activity. Thus, for example, active pumping mechanisms largely control the degree of corneal hydration.

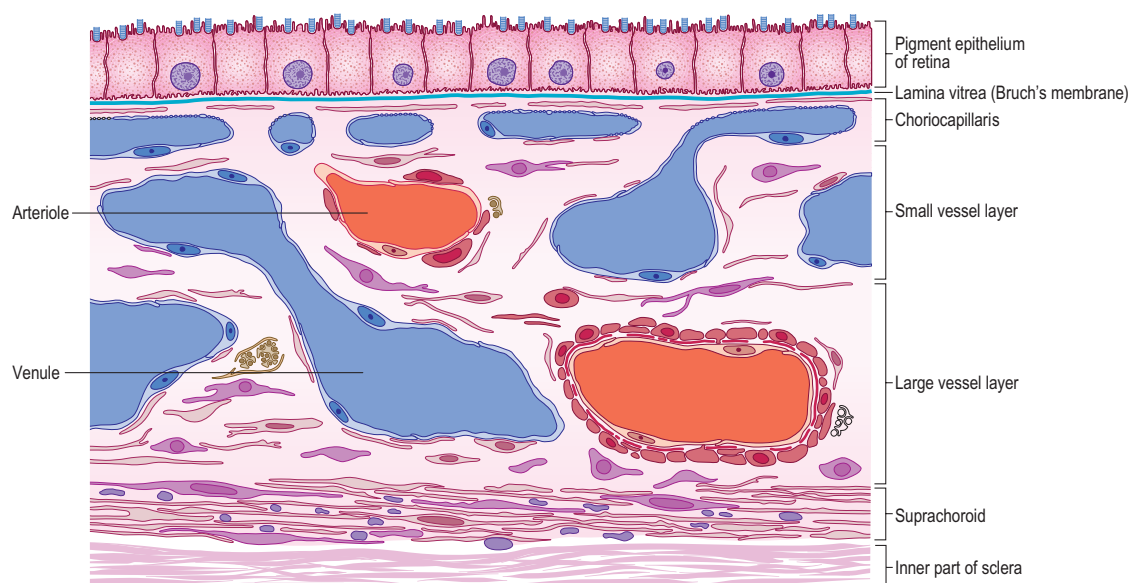


Fig. 42.7 Drawing of a histological section through the choroid close to the posterior pole.

Corneal innervation

The cornea is innervated by numerous branches of the ophthalmic nerve, which either form an anular plexus around the periphery of the cornea, or pass directly from the sclera and enter the corneal stroma radially as 70–80 small groups of fibres. On entering the cornea, the few myelinated nerves lose their myelin sheaths and ramify throughout the anterior corneal stroma in a delicate reticulum, their terminal filaments forming an intricate subepithelial plexus. Axon bundles from this plexus cross the anterior limiting membrane and form a sub-basal plexus from which individual beaded axons pass to more superficial epithelial layers, eventually terminating as free nerve endings. Corneal nerves provide the afferent arm of the blink and lacrimation reflexes (Ch. 41) and may also have a neurotrophic function (Müller et al 2003).

CORNEOSCLERAL JUNCTION (LIMBUS)

The limbus marks the transitional zone between the cornea and sclera (see Fig. 42.3B). Here, the corneal epithelium merges with the epithelium of the conjunctiva, which thickens (up to 12 cells). Bowman's layer terminates and the corneal stromal collagen loses its regularity. As the cornea contains neither blood nor lymphatic vessels, the capillaries of the conjunctiva and episclera end in loops near the limbus. Internally, Descemet's membrane disperses into the fibres of trabecular meshwork and the corneal endothelium is continuous with that covering the trabeculae. The limbus forms a convenient surface landmark for ocular surgery and intravitreal injection. A position 4 mm posterior to the limbus, for example, indicates the pars plana of the ciliary body (orbiculus ciliaris) in adults, through which safe surgical access to the posterior segment of the eye can be achieved.

The corneal epithelium does not possess stem cells and its replacement depends on the centripetal migration of cells from the edges of the cornea. These cells are the progeny of mitotic limbal stem cells located within radially orientated epithelial ingrowths (palisades of Vogt).

UVEA

The vascular tunic of the eye consists of the choroid, ciliary body and iris, which collectively form a continuous structure: the uvea (see Fig. 42.1). The choroid covers the internal scleral surface and extends forwards to the ora serrata. The ciliary body continues forwards from the choroid to the circumference of the iris, which is a circular diaphragm behind the cornea and in front of the lens, forming an almost central aperture, the pupil.

CHOROID

The choroid is a thin, highly vascular, pigmented layer that lines almost five-sixths of the eye posteriorly. Its mean thickness, measured by optical coherence tomography (OCT), is approximately 230 μm (Arora

et al 2012) and is greatest behind the macula. These dimensions are influenced by several factors, including age (Park and Oh 2013), axial length, central corneal thickness and time of day, and vary in pathological conditions such as angle closure.

The choroid is pierced by the optic nerve where it is firmly adherent to the sclera. Elsewhere, its external surface is only loosely connected to the sclera by the suprachoroid layer (lamina fusca). Internally, it is attached to the retinal pigment epithelium, and at the optic disc is continuous with the pia-arachnoid tissues around the optic nerve.

Four layers can be identified in transverse section (Fig. 42.7; see also Fig. 42.22): suprachoroid, vascular stroma, choriocapillaris and lamina vitrea (Bruch's membrane).

Suprachoroid

The suprachoroid covers the external surface of the choroid; it is approximately 30 μm thick and composed of delicate non-vascular lamellae, each one a network of fine collagen and elastic fibres, fibroblasts and melanocytes. Ciliary nerves and long posterior ciliary arteries pass forwards to the anterior uvea in this layer.

Vessel layer (stroma)

Internal to the suprachoroid lies a layer composed mainly of arteries and veins, but also some loose connective tissue containing scattered pigment cells. These melanocytes limit the passage of light through the sclera to the retina. More importantly, like the retinal pigment epithelium, they also absorb light traversing the retina that is not absorbed by the photoreceptors, so preventing internal reflection.

The blood supply of the choroid, as well as the rest of the uvea, is summarized in Figure 42.8. Short posterior ciliary arteries enter the eye through the sclera near the optic disc and supply the posterior choroid. These vessels branch and gradually decrease in size as they approach the retinal border. The choroidal stroma can be divided into layers based on the change in the calibre of these vessels: an outer layer of larger vessels (Haller's layer) and an inner layer of smaller vessels (Sattler's layer), which eventually give rise to the choriocapillaris (see Fig. 42.7). Long posterior ciliary arteries and recurrent branches of anterior ciliary arteries supply the anterior part of the choriocapillaris.

Veins within the choroid converge spirally on to four, or very occasionally more, principal vortex veins. These pierce the sclera behind the equator to reach tributaries of the ophthalmic veins.

The vessels of the choroid have a rich autonomic vasomotor supply.

The blood flow through the choroid is high, a feature probably associated with an intraocular pressure of 15–20 mmHg, which means that a venous pressure above 20 mmHg is required to maintain circulation. The choroidal perfusion rate exceeds that required to supply nutrients and may serve to cool the retina during exposure to bright light.

Choriocapillaris

The choriocapillaris provides nutrients to the avascular outer five layers of the retina and is composed of large (up to 20 μm thick), densely

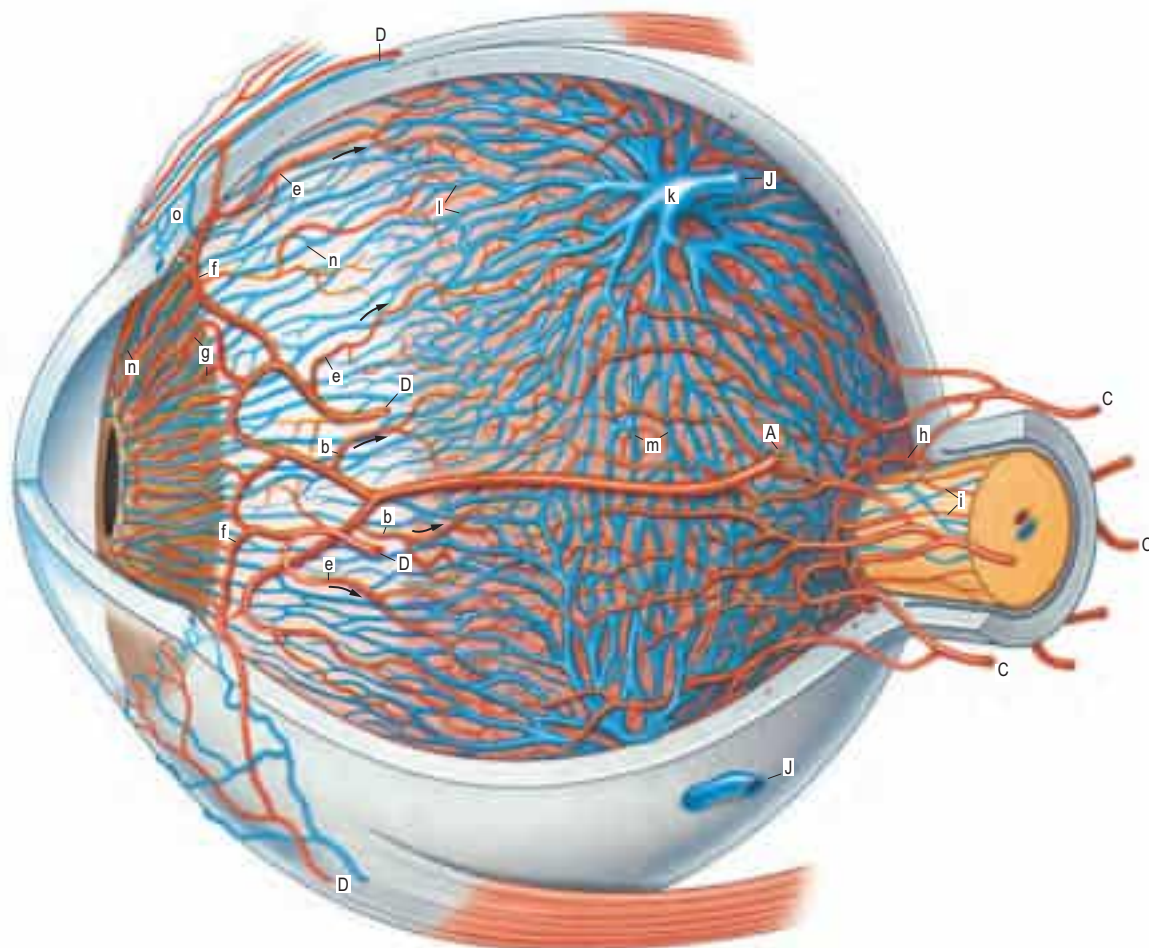


Fig. 42.8 The vasculature of the uveal tract. The long posterior ciliary arteries, one of which is visible (A), branch at the ora serrata (b) and feed the capillaries of the anterior part of the choroid. Short posterior ciliary arteries (C) divide rapidly to form the posterior part of the choriocapillaris. Anterior ciliary arteries (D) send recurrent branches to the choriocapillaris (e) and anterior rami to the major arterial circle (f). Branches from the circle extend into the iris (g) and to the limbus. Branches of the short posterior ciliary arteries (C) form an anastomotic circle (of Zinn) (h) round the optic disc, and twigs from this (i) join an arterial network on the optic nerve. The vortex veins (J) are formed by the junctions (k) of suprachoroidal tributaries (l). Smaller tributaries are also shown (m, n). The veins draining the scleral venous sinus (o) join anterior ciliary veins and vorticoses tributaries. (With permission from Hogan MJ, Alvarado JA, Weddell JE 1971 *Histology of the Human Eye*. Philadelphia: WB Saunders.)

packed, freely anastomosing, fenestrated capillaries. While pericytes occur mainly on the scleral surface, most fenestrations face the retina. The permeability of choriocapillaris is exceptionally high, even compared to other structures invested with fenestrated capillaries.

Lamina vitrea (Bruch's membrane)

The lamina vitrea lies between the choriocapillaris and the retinal pigment epithelium, from both of which it is derived. Under the light microscope, it appears as a homogeneous layer, 2–4 μm in diameter and consists largely of a central elastic fibre mesh, although some authors include flanking internal and external layers of collagen and the basal laminae of the choriocapillaris and retinal pigment epithelium in this structure. Its function is thought to be related to the passage of fluid and solutes from the choroidal capillaries to the retina. In advancing years, extracellular deposits may accumulate in this membrane (drüsen), which impairs the exchange of gases, nutrients and metabolites between the choroidal blood and the outer layers of the retina, potentially contributing to degenerative disease in the photoreceptor layer of the neural retina.

CILIARY BODY

The ciliary body serves to anchor the lens via suspensory ligaments, and by the contraction of its smooth muscle changes the refractive power of the lens (accommodation). Its anterior internal surface is also the source of aqueous humour, while posteriorly its inner surface is contiguous with the vitreous humour and secretes several of its components. The anterior ciliary arteries and the long posterior ciliary arteries meet in the ciliary body, and the major nerves to all the anterior tissues of the eyeball pass through it.

Externally, the ciliary body may be represented by a line that extends from approximately 1.5 mm posterior to the limbus of the cornea (corresponding also to the scleral spur) to a line 7.5–8 mm posterior to this on the temporal side, and 6.5–7 mm on the nasal side. The ciliary body is thus slightly eccentric. It projects posteriorly from the scleral spur, which is its attachment, with a meridional width varying from 5.5 to 6.5 mm. Internally, it exhibits a posteriorly crenated or scalloped periphery, the ora serrata, where it is continuous with the choroid and retina (Figs 42.9–42.10). Anteriorly, it is confluent with the periphery of the iris, and externally, it bounds the iridocorneal angle of the anterior chamber.

In cross-section, the ciliary body is composed of four layers (from internal to external): a double layer of epithelial cells, the stroma, ciliary muscle and a supraciliary layer (see Fig. 42.2).

Meridionally, the ciliary body can be divided into two parts (see Figs 42.2, 42.10). Anteriorly, the ridged pars plicata (corona ciliaris) surrounds the base of the iris and accounts for about one-third of the ciliary body. Posteriorly, the relatively smooth and thin pars plana (orbiculus ciliaris) lies adjacent to the ora serrata, and forms a convenient access point for instruments during vitreoretinal surgery and for intraocular injection. The whole ciliary body is covered by a double epithelium, in which the inner layer is unpigmented, while the outer layer contains melanin. At the ora serrata, the retinal pigment epithelium is continuous with the outer pigmented epithelium of the pars plana, while the neural retina is replaced by inner unpigmented ciliary epithelium (see Fig. 42.24). Anteriorly, this double epithelium continues over the pars plicata until it merges with the double epithelium on the posterior surface of the iris (where the inner layer of cells also accumulates melanin). The anterior pars plicata is ridged meridionally by 70–80 ciliary processes radiating from the base of the iris (see Figs 42.9–42.10). In the young eye, these process are approximately parallel-sided structures, but in the adult, their flanks become less regular and appear thicker. A minor ridge, or ciliary plica, lies in the valley between

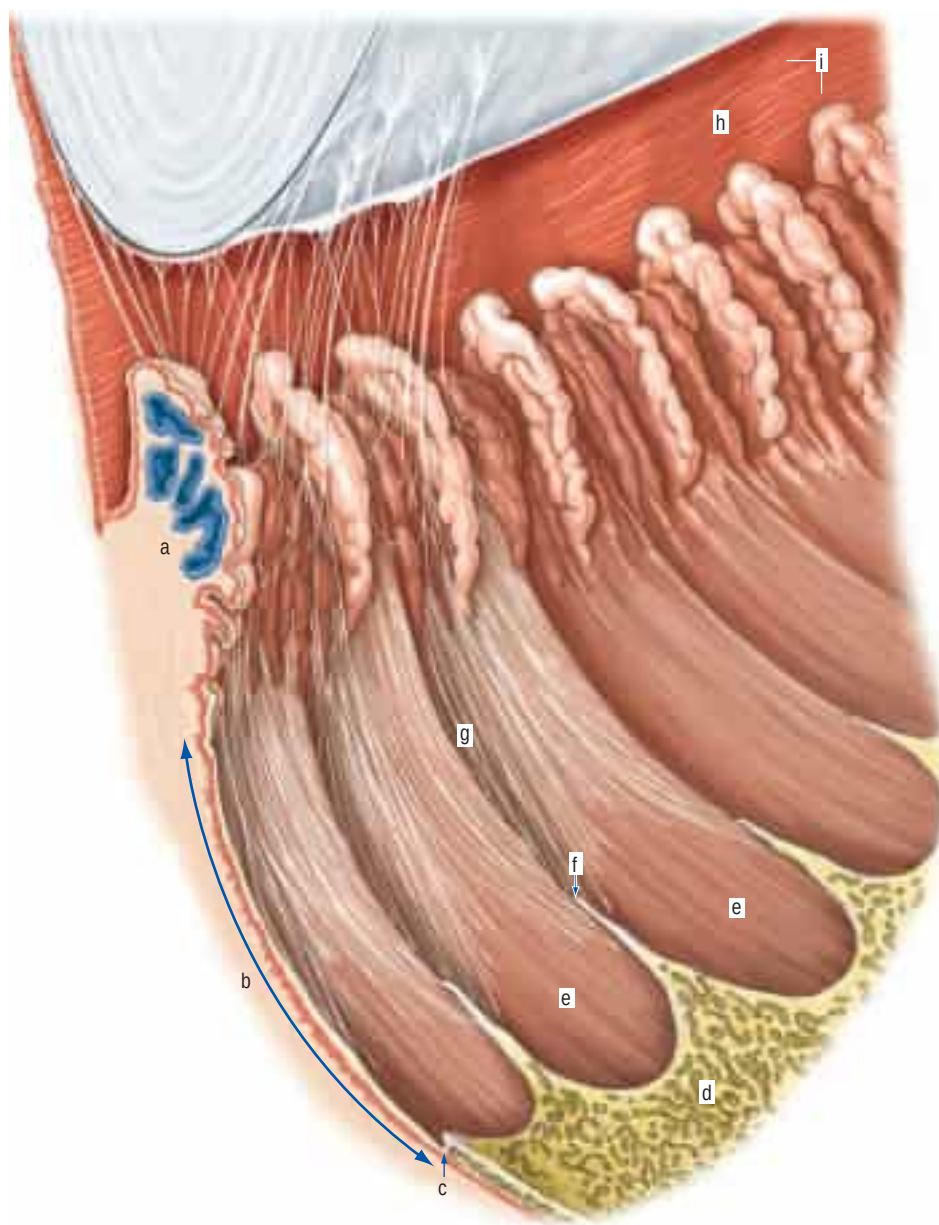


Fig. 42.9 The ciliary region seen from the ocular interior. Above is the periphery of the lens, attached by the fibres of the zonule (suspensory ligament) to the processes of the pars plicata (corona ciliaris) of the ciliary body (a). The pars plana ciliaris (orbiculus ciliaris) (b) has a scalloped boundary, the ora serrata (c), which separates it from the retina (d). Flanking the 'dentate bays' (e) of the ora serrata are the dentate processes (f), with which linear ridges or striae (g) are continuous; the striae extend forwards between the main ciliary processes, providing an attachment for the longer zonular fibres. The posterior aspect of the iris shows radial (h) and circumferential (i) sulci. (With permission from Hogan MJ, Alvarado JA, Weddell JE 1971 *Histology of the Human Eye*. Philadelphia: WB Saunders.)

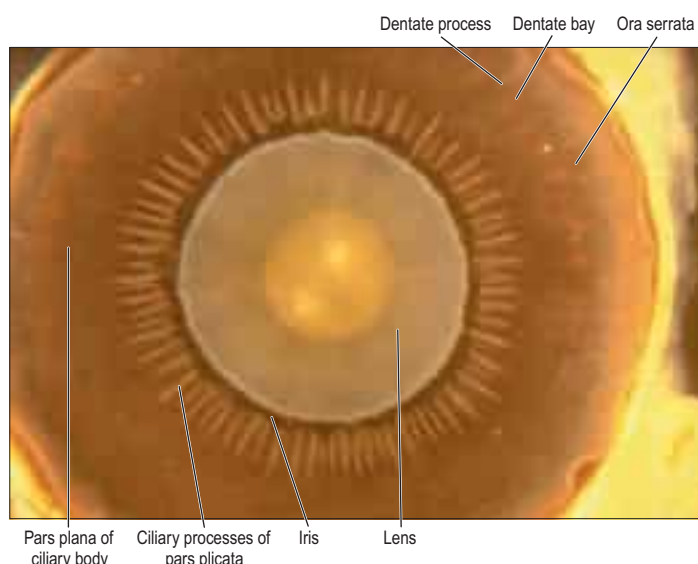


Fig. 42.10 The posterior aspect of the anterior half of the eye, showing the termination of the neural retina at the ora serrata and the ciliary body. The lens has retained sufficient transparency to reveal the iris border; the crenated perimeter of the lens is due to tension imposed by the attached suspensory ligaments (unseen).

most of the processes. The crests of the processes are less pigmented, giving them the appearance of white (or light) striae, from which the name ciliary is derived. The suspensory ligaments attached to the lens extend into the valleys between the ciliary processes and pass beyond them to fuse with the basal lamina of the superficial epithelial layer of pars plana. Their sites of attachment are marked by striae that pass back from the valleys of the pars plicata, across the pars plana, almost as far as the apices of the dentate processes of the ora serrata (see Fig. 42.9).

Ciliary epithelium

The ciliary epithelium is bilaminar, consisting of two layers of epithelial cells that are derived embryonically from the two layers of the optic cup. The superficial layer consists of columnar cells over the pars plana, and cuboidal cells over the ciliary processes of the pars plicata; it becomes irregular and more flattened between the processes. These cells contain little or no pigment. The outer layer of the ciliary epithelium, in contrast, contains cuboidal cells that are loaded with melanin. Normally, the two layers are firmly united but fluid may separate them pathologically. The pigment layer is attached to the stroma of the ciliary body by its basal lamina, which continues back into the basal lamina of the choroid. As a consequence of the invagination of the optic cup during development, a basal lamina covers the free surface of the bilayer, which is continuous with the internal limiting membrane of the retina.

The ciliary epithelium is responsible for the secretion of aqueous humour, while the outer pigmented epithelium additionally

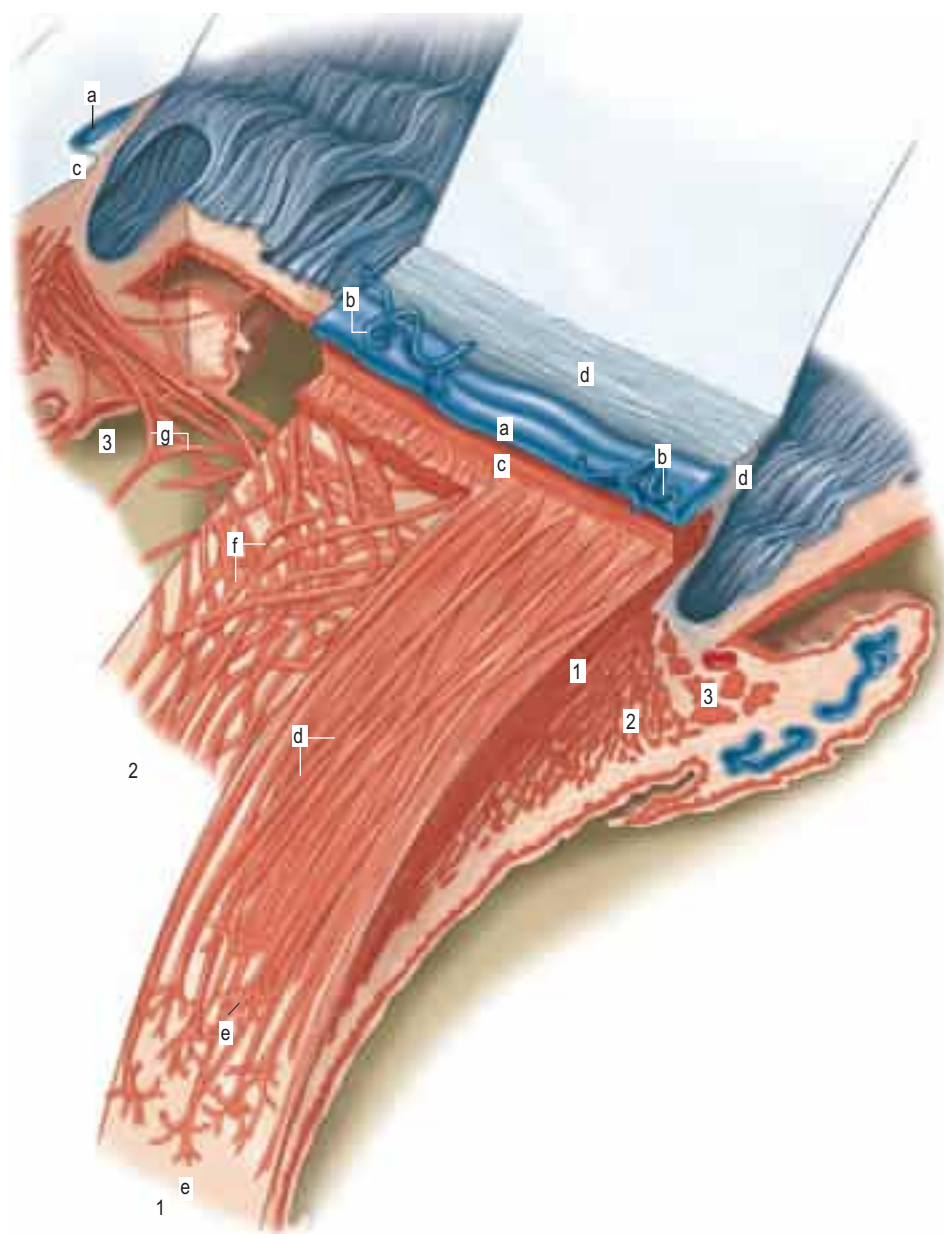


Fig. 42.11 The ciliary muscle and its components. The meridional or longitudinal (1), radial or oblique (2), and circular or sphincteric (3) layers of muscle fibres are displayed by successive removal towards the ocular interior. The cornea and sclera have been removed, leaving the canal of Schlemm (a), collecting venules (b), scleral spur (c) and trabecular meshwork (d). The meridional fibres (1) often display acutely angled junctions (d) and terminate in epichoroidal stars (e). The radial fibres meet at obtuse angles (f) and similar junctions, at even wider angles (g), occur in the circular ciliary muscle. (With permission from Hogan MJ, Alvarado JA, Weddell JE 1971 *Histology of the Human Eye*. Philadelphia: WB Saunders.)

contributes to the eye's 'black box effect', like the retinal pigment epithelium, choroidal pigment and posterior iris epithelium, absorbing stray light to enhance image quality.

Ciliary stroma

The ciliary stroma is composed largely of loose bundles of collagen, which form a considerable mass between the ciliary muscle and overlying processes, and extend into both of them. It also contains numerous larger branches of the ciliary vessels. A dense reticulum of large (up to 35 µm in diameter) fenestrated capillaries is concentrated in the ciliary processes, facilitating the passage of substances from the blood plasma during aqueous production. Anteriorly, near the periphery of the iris, the major arterial circle (see Fig. 42.8) is formed by anastomosis between the anterior ciliary arteries and long posterior ciliary arteries, branches of the ophthalmic artery (see Fig. 41.13). Ciliary veins, also draining the iris, pass posteriorly to join the vortex veins of the choroid.

Ciliary muscle

Ciliary muscle is composed of smooth muscle cells, most of which are attached to the scleral spur and arranged in three different orientations. The outermost fibres are meridional or longitudinal, and pass posteriorly into the stroma of the choroid. The innermost fibres swerve acutely from the spur to run circumferentially as a sphincter near the periphery of the lens. Obliquely interconnecting radial fibres run between these two muscular strata, frequently forming an interweaving lattice (see Fig. 42.2; Fig. 42.11).

Accommodation reflex

At rest, distant objects are focused on the retina in an emmetropic eye. In order to focus closer objects, the dioptric power of the eye has to be increased, which is achieved by increasing the curvature of the lens. At rest, the lens is under tension from the zonular ligaments and hence flattened. On accommodation, the ciliary muscle contracts, moving the ciliary body forwards and inwards towards the optic axis. All parts of the muscle act in concert and tension on the zonular ligaments is relaxed. As the lens is covered by an elastic capsule, once tension on it is released, it assumes a more convex shape suitable for focusing closer objects. The radius of curvature of the anterior lens surface changes most during accommodation. This classic view of accommodation, first proposed in principal by Helmholtz nearly 100 years ago, remains generally accepted, although details of the roles of various ocular structures continue to be debated (Charman 2008).

Information from the retina passing to the visual cortex does not constitute the afferent limb of a simple reflex in the usual sense of the term, but permits the visual areas to assess the clarity of objects in the visual field. Cortical efferent information passes to the pretectal area and thence to the Edinger–Westphal nucleus, which contains preganglionic parasympathetic neurones whose axons travel in the oculomotor nerve and synapse in the ciliary ganglion within the orbit (Fig. 42.12). Postganglionic fibres (short ciliary nerves) innervate the ciliary muscle, causing it to contract. There is also a sparse sympathetic innervation of ciliary muscle, which has a very limited capacity to relax the muscle.

Accommodation is usually accompanied by constriction of the pupil brought about by contraction of the sphincter pupillae, and convergent eye movements caused by contraction of the medial, superior and

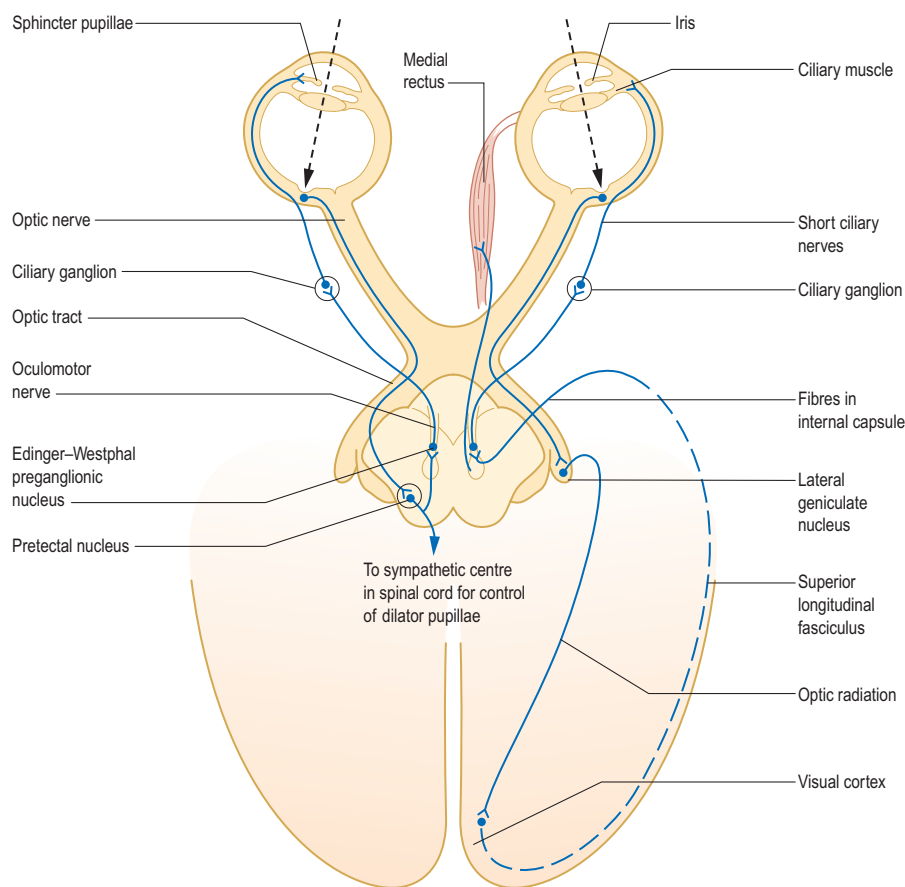


Fig. 42.12 The neural pathways of the pupillary light reflex (left) and the accommodation reflex (right). (From Oxford Textbook of Functional Anatomy, Vol 3 Head and Neck, MacKinnon P, Morris J (eds) 1990. With permission of Oxford University Press.)

inferior recti (all innervated by the oculomotor nerve). This is the 'near triad' that may become disrupted in various diseases.

Supraciliary layer

The thin supraciliary layer separates the sclera from the ciliary muscle and is largely composed of collagen fibres derived from the two layers it divides. It also forms an alternative 'unconventional' route (other than the canal of Schlemm) for the drainage of aqueous humour.

IRIS

The iris is an adjustable diaphragm around a central aperture (slightly medial to true centre), the pupil. It lies between the cornea and lens, and is immersed in aqueous fluid (see Fig. 42.1), partially dividing the anterior segment into an anterior chamber, enclosed by the cornea and iris, and a posterior chamber, situated between the iris and the lens anterior to the vitreous. The efficacy of the iris as a light stop is mainly due to a densely pigmented posterior double epithelium. The pupillary aperture is adjusted by the action of two muscles, dilator and sphincter pupillae.

Seen from the front, the iris is divided into a large ciliary zone adjacent to the ciliary body and a smaller, inner, pupillary zone (Fig. 42.13). The two regions join at the collarette. The anterior surface of the iris often contains large depressions (crypts of Fuchs), and at the pupillary margin, the posterior pigmented epithelium is visible as the pupillary ruff.

In transverse section, several subdivisions of the iris are evident (Fig. 42.14). From anterior to posterior, they are an anterior border layer, the stroma (which contains the sphincter pupillae), and two pigmented epithelial layers, the most anterior of which contains the dilator pupillae. It is thinnest at its root (approximately 200 µm), where the ciliary body is attached, and thickest at the collarette.

Anterior border layer

The anterior surface of the iris is not a distinct epithelium but a modified layer of the iris stroma, formed mainly by an increased number of fibroblasts that constitute a network on the anterior surface with a layer of underlying melanocytes. At the periphery of the iris, the anterior

border layer blends with the trabecular connective tissue of the irido-corneal angle; at the pupillary rim, it meets the epithelium of the posterior surface of the iris.

The colour of the iris is a product of the combined effect of reflection/absorption by the iridial connective tissue and the concentration of pigment in the anterior border layer and, to a lesser extent, in the iris stroma. The distribution of pigment is often irregular, which produces a flecked appearance. When pigment is largely absent, other than in the posterior epithelial layers (the condition at birth), the colour is light blue.

Stroma

The stroma of the iris is formed of fibroblasts, melanocytes and a loose collagenous matrix but no elastic tissue. The intercellular spaces appear to communicate freely with the anterior chamber. A few large, heavily pigmented 'clump cells' may be present; these are phagocytes and contain lysosomes and ingested melanosomes. The stroma contains regional vessels and nerves. An aggregation of smooth muscle cells near the pupillary rim forms an annular contractile sphincter pupillae (see below).

Epithelial layers

The epithelial surface covering the iris posteriorly is a continuation of the bilaminar epithelium of the ciliary body and is formed from the two layers of the optic cup. The pupil, through which this epithelium curves for a short distance on to the anterior surface as the pigment ruff, corresponds to the opening of the optic cup.

The layer of epithelial cells nearest the stroma is, somewhat confusingly, termed the anterior epithelium, although it lies posterior to the stroma. Its cells are pigmented, as are those of the corresponding layer in the ciliary epithelium. They give rise to the dilator pupillae (see below). Vitreal to this stratum is a layer of heavily pigmented cells, the posterior epithelium, which is continuous with the inner non-pigmented layer of the ciliary epithelium. As is readily apparent from *in vivo* scans, the pupillary region of the iris rests on the front surface of the lens (see Fig. 42.3B). Its posterior surface therefore bears numerous radial ridges that facilitate the movement of aqueous humour from the posterior to the anterior chamber. Adjacent epithelial cells are extensively joined by various junctions that ensure the layer can

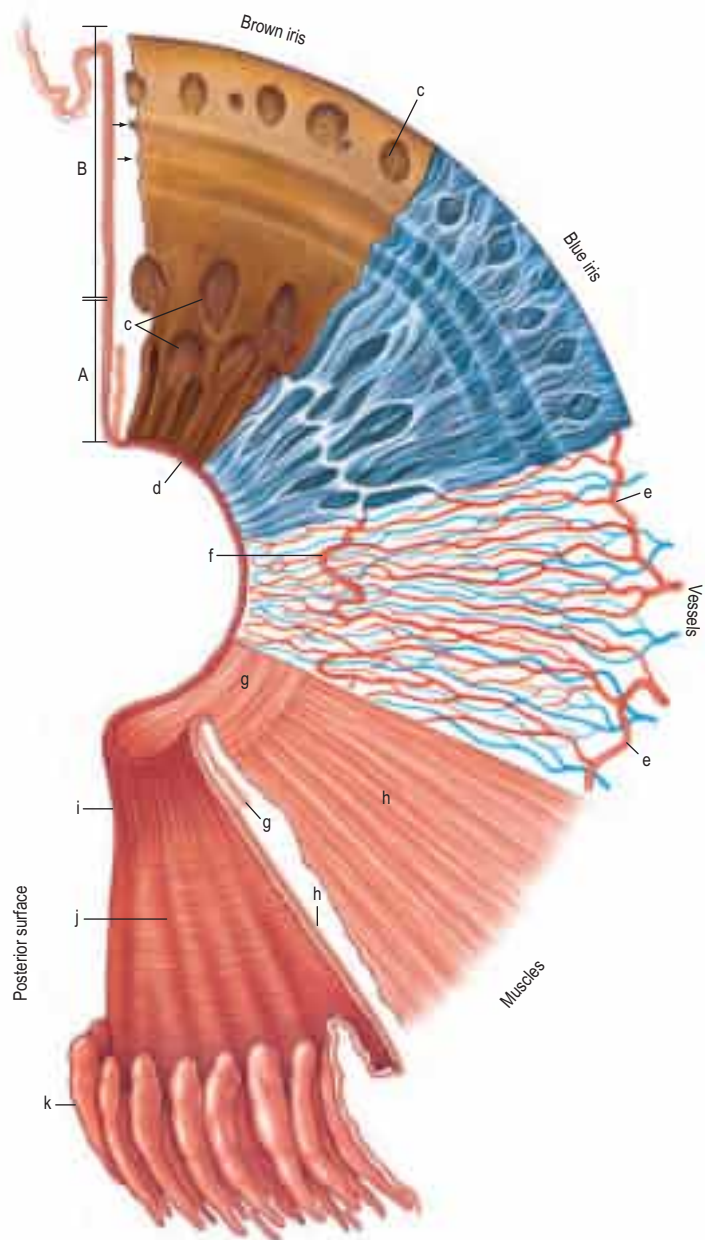


Fig. 42.13 A composite view of the surfaces and internal strata of the iris. In a clockwise direction from above, the pupillary (A) and ciliary (B) zones are shown in successive segments. The first (brown iris) shows the anterior border layer and the openings of crypts (c). In the second segment (blue iris), the layer is much less prominent and the trabeculae of the stroma are more visible. The third segment shows the iridial vessels, including the major arterial circle (e) and the incomplete minor arterial circle (f). The fourth segment shows the muscle stratum, including the sphincter (g) and dilator (h) of the pupil. The everted 'pupillary ruff' of the epithelium on the posterior aspect of the iris (d) appears in all segments. The final segment, folded over for pictorial purposes, depicts this aspect of the iris, showing radial folds (i and j) and the adjoining ciliary processes (k). (With permission from Hogan MJ, Alvarado JA, Weddell JE 1971 *Histology of the Human Eye*. Philadelphia: WB Saunders.)

withstand the excursions of the iris during changes in pupillary size (Fig. 42.15).

Iris muscles

Sphincter pupillae

The sphincter pupillae is a flat anulus of smooth muscle approximately 750 μm wide and 150 μm thick. Its densely packed, fusiform muscle cells are often arranged in small bundles, as in the ciliary muscle, and pass circumferentially around the pupil (see Fig. 42.13). Collagenous connective tissue lies in front of and behind the muscle fibres. It is very dense posteriorly, where it binds the sphincter to the pupillary end of the dilator muscle, and is attached to the epithelial layer at the pupil margin. Small axons, mostly non-myelinated, ramify in the connective tissue between bundles.

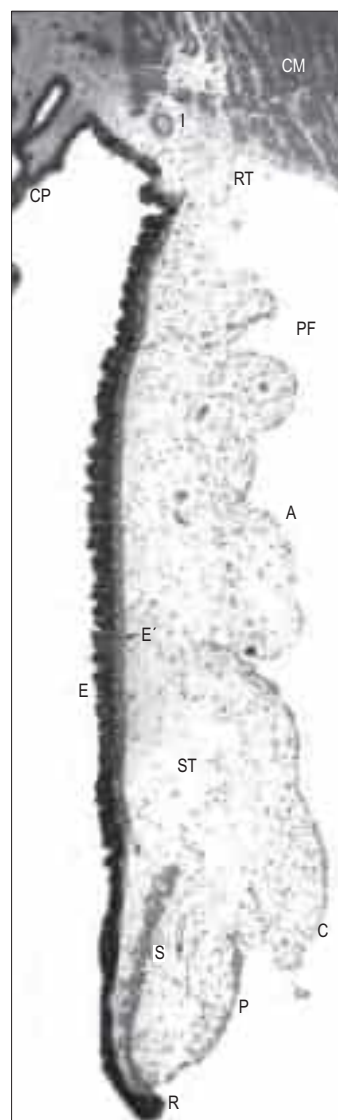


Fig. 42.14 A meridional section of an iris. The posterior surface is lined with a double epithelium (E, E'); the arrow points to the processes of the anterior layer of the epithelium forming the dilator pupillae muscle. The iris is in a contracted state, as shown by the thickened epithelium opposite the dilator muscle and thinning opposite the sphincter, and by the shortness of the pupillary zone. Abbreviations: A, anterior border layer; C, collarette; CM, ciliary muscle; CP, ciliary processes; I, major iridic circle; P, pupillary region of the iris; PF, folds in anterior surface towards the root; R, pupillary ruff; RT, root; S, sphincter pupillae muscle; ST, stroma.

Dilator pupillae

The dilator (dilator) pupillae forms a thin layer that lies immediately anterior to the epithelium of the posterior surface of the iris. Its 'fibres' are the muscular processes of the anterior layer of this epithelium, whose cells are therefore myoepithelial. Myofilaments are present throughout these cells but are more abundant in their fusiform basal muscular processes, which are approximately 4 μm thick, 7 μm wide and 60 μm in length. They form a layer 3–5 elements thick through most of the iris, from its periphery to the outer perimeter of the sphincter, which it slightly overlaps. Here the dilator thins out and sends spurs to blend with the sphincter. Unlike the apical parts of the myoepithelial cells, these have a basal lamina and are joined by gap junctions like those between the sphincteric muscle cells. Small, unmyelinated axons pass between, and terminate on, their muscular processes (see Fig. 42.15).

Innervation of muscles of the iris

The iris is innervated mainly by the long and short ciliary nerves. Short ciliary nerves, which contain parasympathetic, postganglionic, myelinated axons derived from the ciliary ganglion (see Figs 41.17, 42.12), innervate the sphincter pupillae, losing their myelin well before entering the muscle. The dilator is supplied with sympathetic, non-myelinated, postganglionic fibres from the superior cervical ganglion; their routes are less well established. Some go via the ciliary ganglion and reach the eye in the short ciliary nerves, whereas other fibres may travel in the long ciliary nerves, which are branches of the nasociliary nerve. An additional small fraction of nerve endings in the dilator and sphincter muscles have been identified as parasympathetic and sympathetic, respectively, in experimental animal studies, including those on primates. Although ganglion cells have been noted in the iris, the majority of axons are probably postganglionic. They form a plexus around the periphery of the iris, from which fibres extend to innervate the two muscles, the vessels and the anterior border layer; some fibres may be afferent and others are vasomotor.

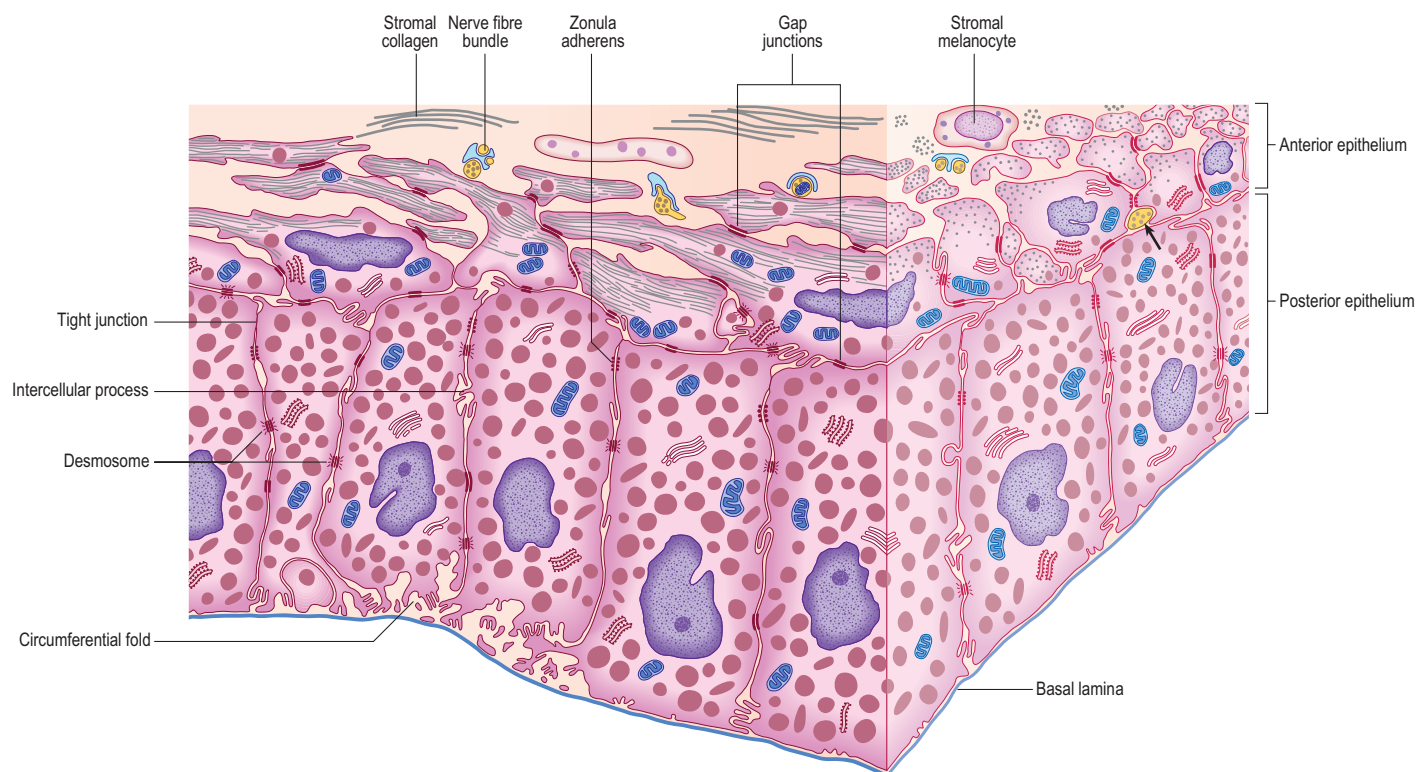


Fig. 42.15 Posterior structures of the iris. The dilator pupillae is shown in transverse section on the right and in longitudinal section on the left (the arrow shows a rarer deeper nerve terminal).

Pupillary light reflex

Pupillary diameter varies from around 2 mm when fully constricted (miosis) in bright light to at least 8 mm when dilated in darkness (mydriasis), and has an even wider range under the influence of drugs. The resulting variation in pupil area (maximally a factor of 16) will obviously affect the amount of light impinging on the retina. However, compared to the total range of illumination within which humans can maintain some degree of vision (approximately 10 log units), this effect, though important, is small. Most mechanisms for dark/light adaptation are retinal, and neural or biochemical in nature. Enhancing visual acuity by restricting light to the centre of the lens, and thereby decreasing the amount of spherical aberration, is at least as important a function of pupillary constriction.

If only one eye is illuminated, the pupil of that eye constricts (direct response), as does the pupil of the contralateral, unilluminated, eye (consensual response). While change in pupillary diameter is usually considered a reflex response to changes in light level, the pupil also constricts on viewing near objects (as part of the near triad; p. 693) and in response to painful stimuli; it can also be influenced to some degree by more complex cortical factors.

In pupillary constriction, light acting on both traditional retinal photoreceptors (rods and cones) and on intrinsically photosensitive retinal ganglion cells gives rise to activity in retinal ganglion cells. This activity is conducted along the optic nerve, through the optic chiasma and along the optic tract. Although the majority of tract fibres end in the lateral geniculate nucleus of the thalamus, a small number leave the optic tract before it reaches the thalamus, at the superior brachium, and synapse in the olivary pretectal nucleus. The information is relayed from the pretectal nucleus by short neurones that synapse bilaterally on preganglionic parasympathetic neurones in the Edinger–Westphal nucleus (in the oculomotor nerve complex in the rostral midbrain). Efferent impulses pass along parasympathetic fibres carried by the oculomotor nerve to the orbit, where they synapse in the ciliary ganglion. Postganglionic fibres travel in the short ciliary nerves to the sphincter pupillae, which reduces the size of the pupil when it contracts (see Fig. 42.12).

Pupillary dilation is brought about by lessening the parasympathetic drive to the sphincter (see above) and by sympathetic activation of the dilator. Sympathetic preganglionic fibres arise from neurones in the lateral column of the first and second thoracic segments, and pass via the sympathetic trunk to the superior cervical ganglion. Postganglionic neurones travel up the neck next to the internal carotid artery as the internal carotid nerve; at the level of the cavernous sinus, the nerve breaks up to form an interweaving network of fibres, the carotid plexus,

around the carotid artery. Some of the axons from the plexus form the sympathetic root of the ciliary ganglion, passing through the ciliary ganglion without synapsing; mostly travelling in the short ciliary nerves, they innervate the dilator.

Vascular supply of the iris

The iris receives its blood supply from the long posterior and anterior ciliary arteries (see Figs 41.13, 42.8). On reaching the attached margin of the iris, both long ciliary arteries divide into an upper and a lower branch. The branches anastomose with the corresponding contralateral arteries, and with the anterior ciliary arteries, to form the major arterial (iridic) circle (circulus arteriosus major) at the base of the iris (see Figs 42.13–42.14). Vessels converge from this circle towards the free margin of the iris, where they form loops and become veins. At the level of the collarette, arteries and veins anastomose to form an incomplete minor iridic circle (circulus arteriosus minor). The smaller arteries and veins are very similar in their structure and are often slightly helical, which allows them to adapt to changes in iridial shape as the pupil varies in size. All of the vessels, including the capillaries, have a non-fenestrated endothelium and a prominent, often thick, basal lamina. There is no elastic lamina in the arteries or veins, and there are few smooth muscle cells, especially in the veins. Connective tissue in the tunica media is loose, whereas the adventitia is remarkably dense and collagenous, so that it appears to form almost a separate tube.

LENS AND HUMOURS

The cornea, aqueous humour, lens and vitreous body, often collectively termed the ocular media, serve to form an image on the retina by transmitting and refracting light. Additionally, the aqueous provides nutrients to the avascular cornea and lens, and removes their metabolic waste, as well as generating the intraocular pressure that maintains the shape of the eye.

AQUEOUS HUMOUR

Aqueous humour is derived from the plasma within the fenestrated capillaries of the ciliary processes. The major component of aqueous, like plasma, is water and the composition of the two fluids is broadly similar, although they do differ in the concentration of some electrolytes and organic solutes. In the interests of optical clarity, the

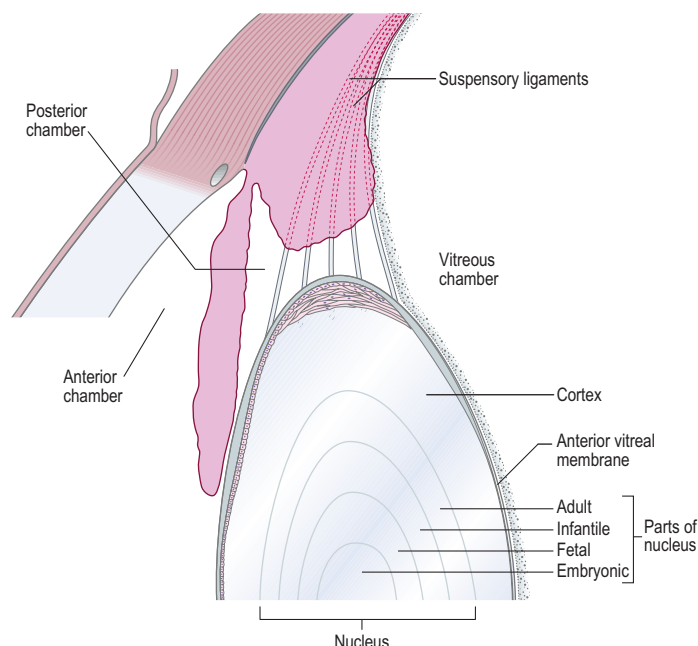


Fig. 42.16 A sagittal section through part of the lens and related structures. The cortex is distinguished from the nucleus, which is composed of embryonic, fetal, infantile and adult parts. The capsule is drawn in blue, exaggerated in thickness 100 times, and is based on observations from a 35-year-old.

blood–aqueous barrier also ensures a very low concentration of protein in the aqueous (generally less than 1% of the level in plasma). Inflammation of the anterior uvea can lead to a breakdown of this barrier and the presence of protein in the aqueous, resulting in light scatter that is manifest clinically as ‘flare’.

The aqueous is actively secreted into the posterior chamber by the epithelium overlying the ciliary processes. It passes around the equator of the lens and flows through the pupil into the anterior chamber, where it circulates before being drained at the iridocorneal angle. Most aqueous is drained from the eye through the trabecular meshwork into the canal of Schlemm; from here, it drains into episcleral veins. However, some exits through the ciliary muscle into the supraciliary and suprachoroidal spaces (uveoscleral pathway).

Any interference with the drainage of aqueous into the canal of Schlemm increases intraocular pressure and leads to glaucoma. Glaucoma is either ‘primary’, or secondary to a specific anomaly or disease of the eye. Primary glaucoma can be either ‘closed angle’, where the filtration angle is narrowed by the proximity of the root of the iris to the cornea, or, more commonly ‘open angle’, where aqueous access to angle tissues is unimpeded. In the most common form of primary open angle glaucoma, pathological changes within the trabecular meshwork reduce the facility of aqueous drainage, thus raising intraocular pressure. Sustained raised pressure leads to progressive defects in the visual field, because of either direct mechanical damage to retinal ganglion cell axons (particularly at the optic nerve head), or impairment of the optic nerve head blood supply, or both.

LENS

The lens is a transparent, encapsulated, biconvex body bathed in aqueous humour, which serves to adjust the focus of the eye. Posteriorly, it contacts the hyaloid fossa of the vitreous body. Anteriorly, it forms a ring of contact with the posterior border of the pupillary region of the iris but, further away from the axis of the lens, the gap between the lens and iris increases to form the posterior chamber of the eye (see Fig. 42.1; Fig. 42.16). The lens is encircled by the ciliary processes, and is attached to them by the zonular fibres, which issue mainly from the pars plana of the ciliary body. Collectively, the fibres form the zonule, which holds the lens in place and transmits the forces that stretch the lens.

The anterior convexity of the lens is less steep than that of the posterior surface. The central points of these surfaces are the anterior and posterior poles of the lens and a line connecting them is its axis. The marginal circumference of the lens is its equator.

At birth, the lens is colourless and transmits all wavelengths from the infra-red to the near-ultraviolet well. However, throughout life, the



Fig. 42.17 A section through the anterior layers of the lens. The thin capsule covers the single row of epithelial cells (arrow), and the lens substance is composed of regularly stacked younger fibres and more densely stained and complex older deeper fibres.

amount of short-wave radiation transmitted diminishes until, in old age, the lens takes on an amber tinge as it absorbs visible short-wave radiation, decreasing blue sensitivity in older people. In cataract, the lens gradually becomes opaque.

The dimensions of the lens are optically and clinically important, but change with age as a consequence of continuous growth. Its equatorial diameter at birth is 6.5 mm, increasing rapidly at first, then more slowly to 9.0 mm at 15 years of age, and even more gradually to reach 9.5 mm in the ninth decade. Its axial dimension increases from 3.5–4.0 mm at birth to 4.75–5.0 mm at age 95. The lens flattens during the first decade of life (Mutti et al 1998). Average adult radii of the anterior and posterior surfaces of the lens are 10 mm and 6 mm, respectively. These reduce in later life, the anterior surface increasing most in curvature as the lens thickens, so that, in old age, the lens pushes the iris forwards slightly, predisposing the elderly eye to angle closure glaucoma.

The lens consists of three layers (Fig. 42.17). Its bulk is composed of elongated cells (lens fibres), which anteriorly, as far as the equator, are covered by a single layer of epithelial cells, and the whole is surrounded by the lens capsule. The lens is avascular and devoid of nerve fibres or other structures that might affect its transparency. Its surface forms a very effective barrier against invasion by cells or elements of the immune system, and so creates an immunologically sequestered environment. It is unique because it retains all the cells formed throughout its life.

Lens capsule

The lens capsule is a basement membrane that covers the surface of the lens. It consists of various classes of collagen fibre (I, III and IV), as well as a range of glycosaminoglycans and glycoproteins. It is derived from the anterior lens epithelial cells and their fetal precursors, and is firmly attached to the epithelium anteriorly and to lens fibres posteriorly. The capsule is elastic, a property it owes to the arrangement of its collagen, and which is central to the process of accommodation. Capsule thickness varies with position and age. One view is that, in adults, it is thicker at the anterior pole (12–14 μm) and equator than at the posterior pole, with a band of increased thickness between the pole and equator (anteriorly, 20–24 μm) (see Fig. 42.16). Zonular fibres insert into the capsule at the equator. They are composed of thin (4–7 nm) fibrils with hollow centres, and resemble fibrils associated with elastic connective tissue.

Lens epithelium

The anterior surface of the lens between the outer capsule and the underlying lens fibres is covered by a layer of simple, roughly cuboidal (10 μm high and 13 μm wide), epithelial cells that, in surface view, are polygonal (see Fig. 42.17). These cells differentiate into lens fibres; they undergo mitosis at a germinative zone just anterior to the equator and are displaced towards the equator, where they synthesize characteristic lens fibre proteins and undergo extreme elongation. As other cells follow suit, the earlier cells come to occupy a deeper position within the lens.

Lens fibres

The inwardly displaced epithelial cells elongate further in both an anterior and a posterior direction, becoming up to 12 mm long, depending on age and position in the lens. Fibres near the surface at the equator are nucleated; the nuclei form a short, S-shaped bow, which extends inwards from the surface. The deeper fibres lose their nuclei and most other organelles.

Variations in lens fibre structure and composition make it possible to distinguish a softer cortical zone, made up of younger fibres, and a firmer central part, representing the older nucleus. The nucleus can be further subdivided into layers representing the age at which the fibres within them were formed.

In cross-section, individual fibres are flattened hexagons measuring approximately 10 μm by 2 μm . They are tightly packed, and fibres are firmly attached to their neighbours by a variety of mechanical junctions (Fig. 42.18). Lens fibres are also in contact through desmosomes and numerous gap junctions.

All lens fibres cross a plane passing through the equator and terminate on both the anterior and posterior lens surfaces at lens 'sutures'. These radiate out from the poles towards the equator and represent lines of linearly registered, interlocking junctions between terminating lens fibres. In fetuses, the sutures on the anterior surface of the lens form a triradiate pattern centred on the anterior pole resembling the limbs of an upright letter Y (Fig. 42.19A). Posteriorly, the sutural configuration is similar but inverted. The sutures increase in number and complexity as a consequence of lens growth and other changes in the arrangement of lens fibres (Fig. 42.19B). Fibres that start near the central axis of the lens anteriorly terminate posteriorly on a suture near the periphery, and *vice versa*.

Lens fibres contain crystallins, proteins that are responsible for the transparency and refractive properties, and for much of the elasticity, of

the lens. At least three varieties coexist – α , β and γ , their relative proportions changing throughout life. They occur in very high concentrations, and form up to 60% of the lens fibre mass. Variations in their concentration in different parts of the lens give rise to regional differences in refractive index, correcting for the spherical and chromatic aberrations which might otherwise occur in a homogeneous lens.

Ocular refraction

The cornea and humours have a refractive index close to that of water, but the tear film covering the curved corneal surface is in contact with air and therefore approximately two-thirds (approximately 40 dioptres) of the refractive power of the eye is effected here. The lens has a greater refractive index than the adjacent media, varying from 1.386 at its periphery to 1.406 at its core, and contributes the remaining approximately 20 dioptres of the power of the relaxed eye. The main value of the lens is its ability to vary its dioptric power during the process of accommodation. Accommodation allows an increase in refraction of 12 dioptres in youth but this decreases with age, being halved at 40 years and reduced to 1 dioptre or less at 60 years (presbyopia).

Disorders of refraction

A relaxed eye is said to be emmetropic when the refracting structures are so related to its length that the retina receives a focused image of a distant object. Although the majority of eyes are emmetropic, a large

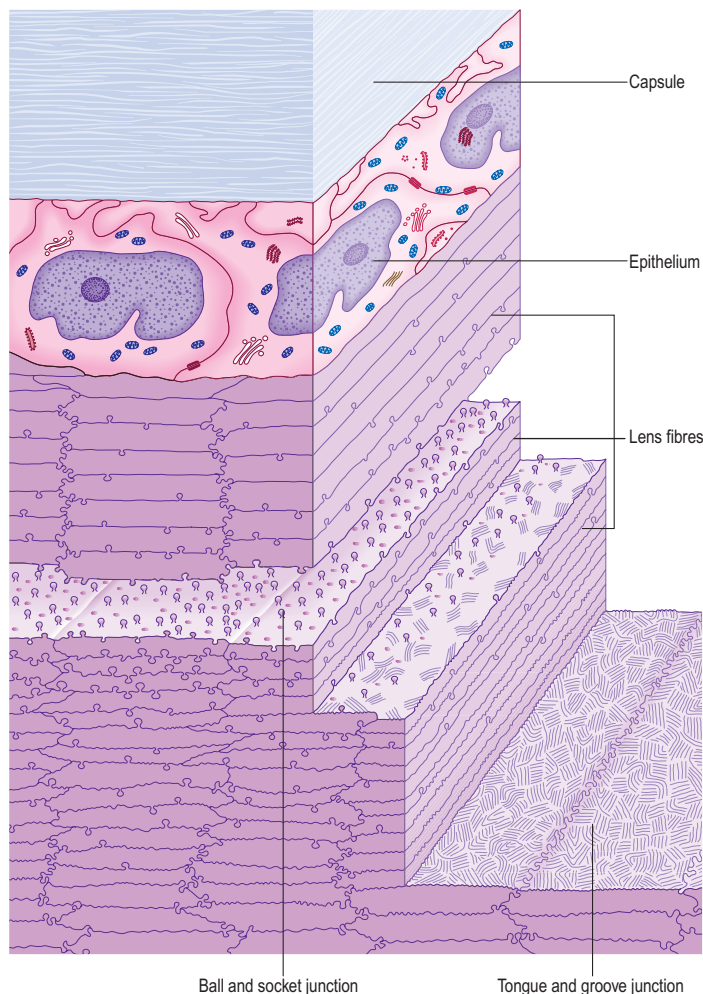


Fig. 42.18 The mechanical junctions between lens fibres. There are no junctions immediately below the epithelium. Ball and socket junctions develop in deeper layers; these are subsequently eliminated towards the nucleus, where tongue and groove junctions gradually form. Angle joints are present at most levels.

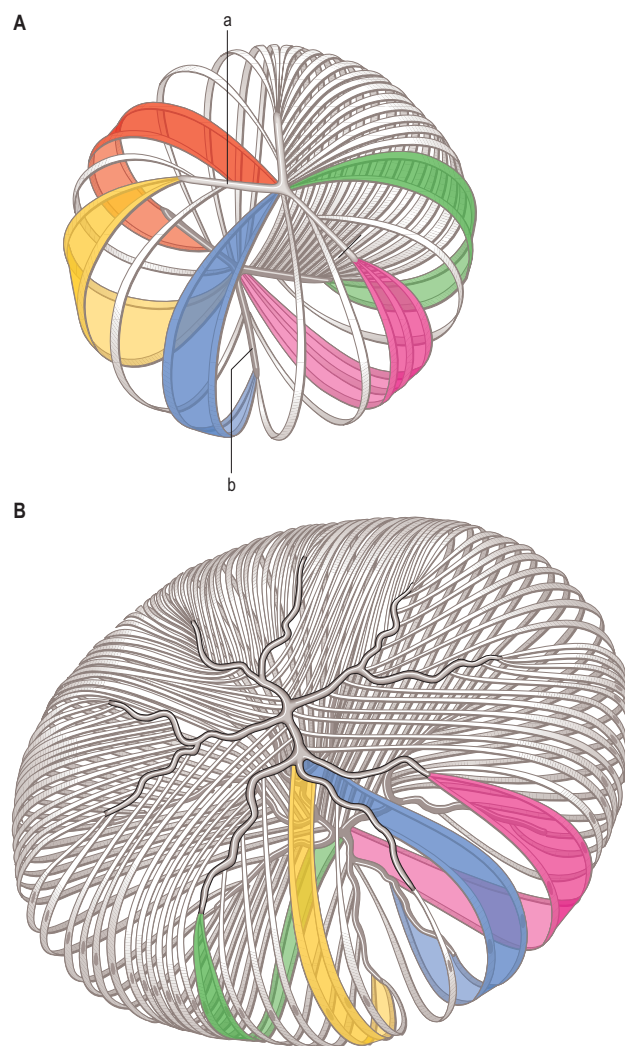


Fig. 42.19 The structure of the fetal (A) and adult (B) human lens, showing the major details of arrangement of the lens fibres. The anterior (a) and posterior (b) triradiate sutures are shown in the fetal lens. Fibres pass from the apex of an arm of one suture to the angle between two arms at the opposite pole, as shown in the coloured segments. Intermediate fibres show the same reciprocal behaviour. The suture pattern becomes much more complex as successive strata are added to the exterior of the growing lens, and the original arms of each triradiate suture show secondary and tertiary dichotomous branchings. (With permission from Hogan MJ, Alvarado JA, Weddell JE 1971 *Histology of the Human Eye*. Philadelphia: WB Saunders.)

minority have errors of refraction or ametropia that can take three different forms. In myopia, the eye is too long for its refractive power and distant objects are focused in front of the retina when relaxed; closer objects will be in focus and, consequently, the eye is said to be 'short-sighted'. Conversely, in hyperopia (long sight), the eye is too short for its refractive power and, when it is relaxed, distant objects are focused behind the retina. In astigmatism, the refractive power of the eye is not the same in different meridians, which are approximately 90° apart in regular astigmatism.

Normally, emmetropia is assured during development both by the active control of the sclera's biomechanical and biochemical properties and changes in the optical features of the eye's refractive elements, a process known as emmetropization (Wallman and Winawer 2004). A resting focal plane behind the retina results in scleral growth causing axial elongation of the globe until the focused image and the position of the retina are coincident. On the other hand, light focused in front of the retina retards scleral growth, reducing axial length. In ametropia, this process fails.

The causes of refractive errors such as myopia are both genetic and environmental. Not only have several candidate genes been identified, but also factors such as increased near work and lack of outdoor activity have all been linked to myopia (Wallman and Winawer 2004, Flitcroft 2013). This has led to a recent increase in the incidence of myopia to epidemic proportions, with a prevalence of over 80% in the adult population of some East Asian cities (Rose et al 2008). Fortunately, errors of refraction are amenable to correction using spectacle or contact lenses and by various forms of refractive surgery.

As noted above, the ability to change the power of the lens through accommodation diminishes during the fifth decade to an extent that neither the corrected ametropia nor the emmetrope is able to focus near objects clearly, and reading spectacles become necessary. Many factors could potentially cause such loss of accommodation, but it seems likely that the main cause is reduced lens elasticity with age. This is offset to a very limited extent by the reduction of the pupil aperture with age, which increases the depth of focus but at the cost of creating the further problem of requiring greater illumination.

Other errors of refraction are the concomitants of eye disease, especially those that affect the cornea. Corneal curvature, for example, may be sufficiently altered as a residual defect of past disease to cause irregular astigmatism. In keratoconus, the cornea is thinned and steepened centrally, distorting the refracting surface. A dislocated lens, caused, for example, by Marfan's syndrome, also disrupts the refractive status of the eye.

VITREOUS HUMOUR

The vitreous body occupies about four-fifths of the eyeball. Posteriorly, it is in contact with the retina, while further forwards it abuts the ciliary body, zonule and lens. Its anterior surface is hollowed into a deep concavity, the hyaloid fossa, fitting the shape of lens (see Fig. 42.1). It is colourless, consisting of approximately 99% water, but is not entirely structureless. At its periphery, it has a gel-like consistency (100–300 µm thick); nearer the centre, it contains a more liquid zone. Hyaluronan, in the form of long glycosaminoglycan chains, fills the whole vitreous. In addition, the peripheral gel or cortex contains a random loose network of type II collagen fibrils, which are occasionally grouped into fibres. The cortex also contains scattered cells, the hyalocytes, which possess the characteristics of mononuclear phagocytes and may contribute to the production of hyaluronan. While they are normally in a resting state, they have the capacity to be actively phagocytic in inflammatory conditions. Hyalocytes are not present in the cortex bordering the lens.

The liquid vitreous is absent at birth, appears first at 4 or 5 years, and increases to occupy half the vitreous space by the seventh decade. Vitreous liquefaction results in an increased incidence of posterior vitreous detachment and associated floaters in the elderly. The cortex is most dense at the pars plana of the ciliary body adjacent to the ora serrata, where attachment is strongest, and this is often referred to as the base of the vitreous. Apart from the vitreous base, the vitreous also has a firm (peripapillary) attachment at the edge of the optic disc. This adherence of the vitreous to the retina can result in traction on the retina if the vitreous shrinks, such as occurs in old age, resulting in macular holes or peripheral breaks, possibly leading to retinal detachment.

A narrow hyaloid canal runs from the optic nerve head to the central posterior surface of the lens (see Fig. 42.1). In the fetus, this contains the hyaloid artery, which normally disappears about 6 weeks before birth. The canal persists in adult life as a very delicate fibrous structure and is of no functional importance.

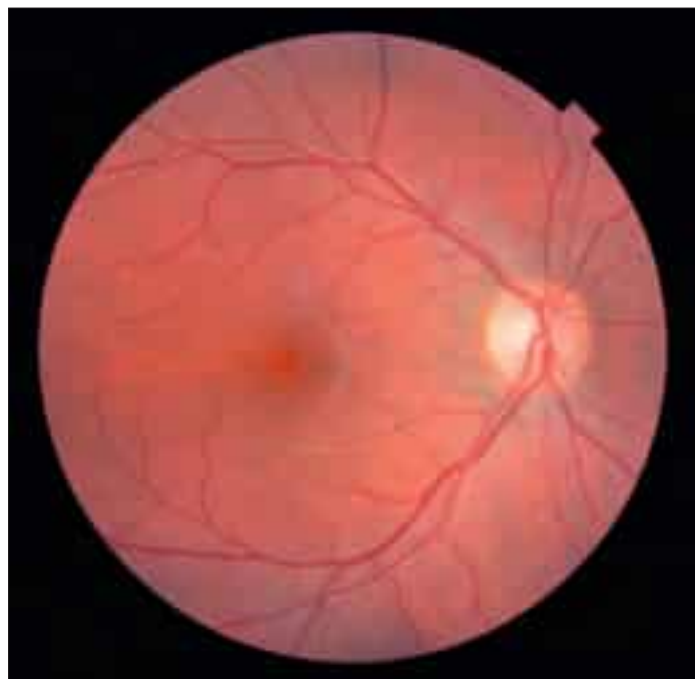


Fig. 42.20A A fundus photograph of the right eye of a 19-year-old Caucasian female. The central retinal vessels are seen emanating from the optic disc. Retinal arteries are narrower and lighter in colour, and generally are vitreal to the veins. The avascular centre of the macular region, with its associated macular pigment, can be seen temporal to the disc.

RETINA

The retina is a thin sheet of cells, ranging from less than 100 µm at its edge to a maximum of around 300 µm at the foveal rim. It lines the inner posterior surface of the eyeball, sandwiched between the choroid externally and the vitreous body internally, and terminates anteriorly at the ora serrata (see Fig. 42.1).

When it is viewed with an ophthalmoscope to show the fundus oculi, the most prominent feature is the blood vessels emanating from and entering the optic disc (Fig. 42.20A). Centred temporal and inferior to the disc lies the 'central retina' or macula (diameter 5–6 mm), the middle of which is composed of the fovea and foveola, and easily identified with an ophthalmoscope as an avascular area with a yellow tinge (see Fig. 42.20A). The lack of blood vessels at the foveola is even more apparent in a fluorescein angiogram. The peripheral retina lies outside the central retina (Fig. 42.20B).

MICROSTRUCTURE

The retina is composed of a variety of epithelial, neural and glial cell types, whose distribution conventionally divides it into 10 layers (Fig. 42.21). These are apparent in histological sections (Fig. 42.22), but can also be seen *in vivo* using optical coherence tomography, an imaging technique that uses backscattered light to visualize layers by differences in their optical scattering properties (Fig. 42.23). Embryologically, the retina is derived from the two layers of the invaginated optic vesicle. The outer layer becomes a stratum of cuboidal pigment cells that separates the choroid from the neural retina, and therefore forms the outermost layer of the retina: the retinal pigment epithelium (layer 1). The other nine strata of the retina develop from the inner layer of the optic vesicle and form the neural retina.

The outermost layer of the neural retina contains the light-sensitive parts of the photoreceptors, which convert the optical image into neural activity. From the photoreceptors, neural activity flows radially to bipolar and ganglion cells, and laterally via horizontal cells in the outer retina and amacrine cells in the inner retina. Photoreceptors synaptically contact each other and bipolar and horizontal cells in the outer plexiform layer (layer 5), while bipolar, amacrine and ganglion cells synapse in the inner plexiform layer (layer 7). The axons of ganglion cells run towards the optic disc in the nerve fibre layer (layer 9), where they leave the retina as the optic nerve, which transmits the retinal output to the visual areas of the brain where visual processing is completed. Although most neural activity flows from the



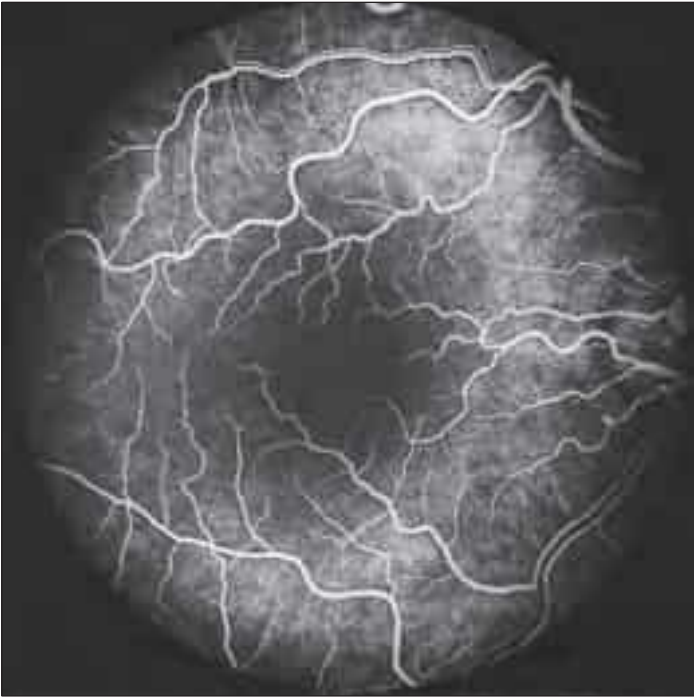


Fig. 42.20B Fluorescein angiogram showing the macular region of a right eye. The main macular vessels are approaching from the right. The subject was an elderly person with considerable macular pigmentation, which masks fluorescence from the choroidal circulation.

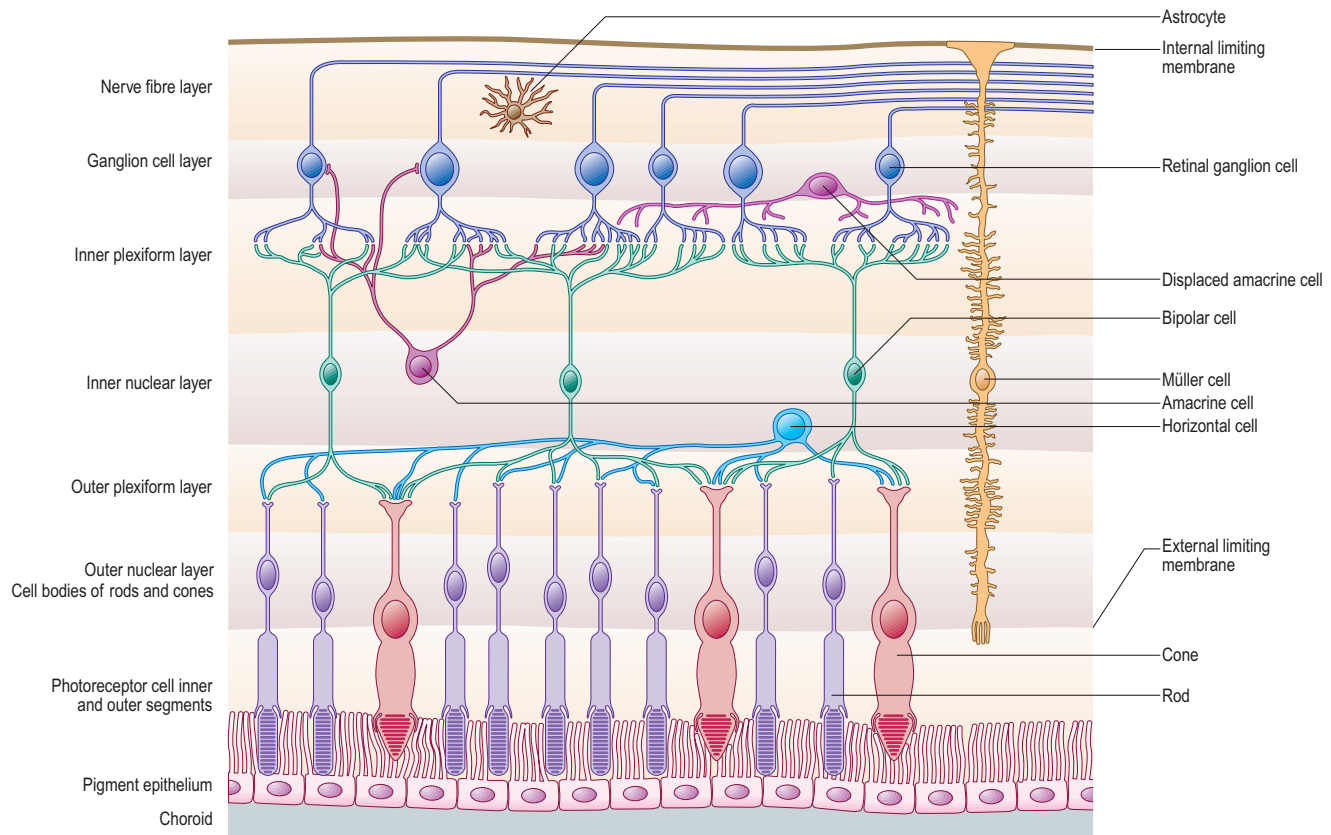


Fig. 42.21 Neural cells whose cell bodies and interconnections account for the layered appearance of the retina in histological section (compare with Fig. 42.22). Also shown are the two principal types of neuroglial cell in the retina (although microglia are also present, they are not shown).

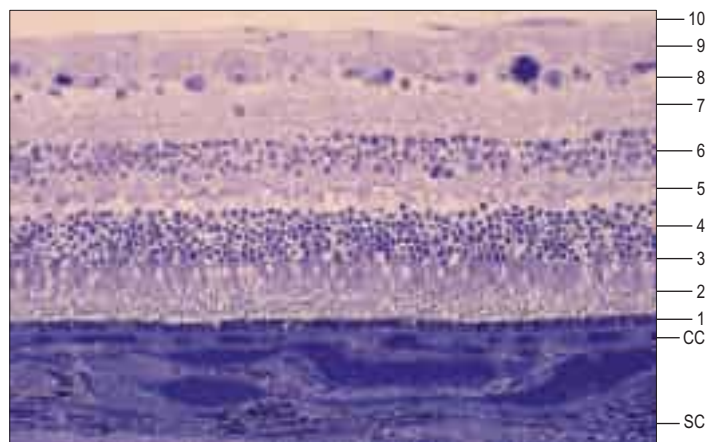


Fig. 42.22 A transverse section of the retina and choroid. The 10 layers of the retina are shown. Key: 1, pigment epithelial layer; 2, rod and cone layer; 3, external limiting membrane; 4, outer nuclear layer; 5, outer plexiform layer; 6, inner nuclear layer; 7, inner plexiform layer; 8, ganglion cell layer; 9, nerve fibre layer; 10, internal limiting membrane. Abbreviations: CC, choriocapillaris; SC, suprachoroid.

photoreceptors towards the brain, some information flow occurs in the opposite direction via centripetal fibres in the optic nerve and interplexiform cells in the retina that connect the inner and outer plexiform layers.

The classic 10-layered appearance of the retina is absent in the optic nerve head, the fovea and foveola, and the ora serrata. At the optic nerve head, the axons of the retinal ganglion cells leave the retina to form the optic nerve and all the other neural cell types are missing. At the fovea and foveola, the inner five layers of the retina are 'pushed aside'. At the ora serrata, where the retina borders the ciliary body (see Fig. 42.10), the retinal pigment epithelium merges with the outer pigmented epithelium of the ciliary body, while the neural retina borders the inner unpigmented ciliary epithelium; the retina is thinnest at this point. The normal layered arrangement of the neural retina approaching the ora serrata is frequently disrupted by cysts in older individuals (Fig. 42.24).

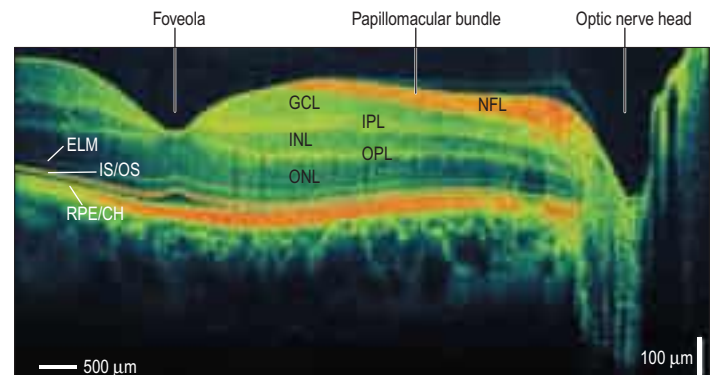


Fig. 42.23 A high-definition optical coherence tomography (OCT) *in vivo* image of the human retina. The image has approximately 2 μm axial resolution, is 8 mm long and consists of 10,000 axial scans. The image is expanded in the vertical direction to permit better visualization of retinal layers. Abbreviations: ELM, external limiting membrane; GCL, ganglion cell layer; INL, inner nuclear layer; IPL, inner plexiform layer; IS/OS, boundary between the photoreceptor inner and outer segments; NFL, nerve fibre layer; ONL, outer nuclear layer; OPL, outer plexiform layer; RPE/CH, retinal pigment epithelium and choriocapillaris. (Courtesy of Professor James Fujimoto, Department of Electrical Engineering and Computer Science, MIT, Boston, USA.)

Cells of the retina

Retinal pigment epithelium

The retinal pigment epithelium is composed of approximately cuboidal cells that form a single continuous layer extending from the periphery of the optic disc to the ora serrata, where it continues as the outer ciliary epithelium. The cells are flat in radial section and hexagonal or pentagonal in surface view, and number 4–6 million in the human retina. Their cytoplasm contains numerous melanosomes. Apically (towards the rods and cones), the cells bear long (5–7 μm) microvilli that contact, or project between, the outer segments of rods and cones. The tips of rod outer segments are deeply inserted into invaginations in the apical membrane of the retinal pigment epithelium. The different

embryological origins of the retinal pigment epithelium and neural retina mean that the attachments between these two layers are unsupported by junctional complexes; the neural retina and retinal pigment epithelium are therefore easily parted (retinal detachment) due to trauma or disease.

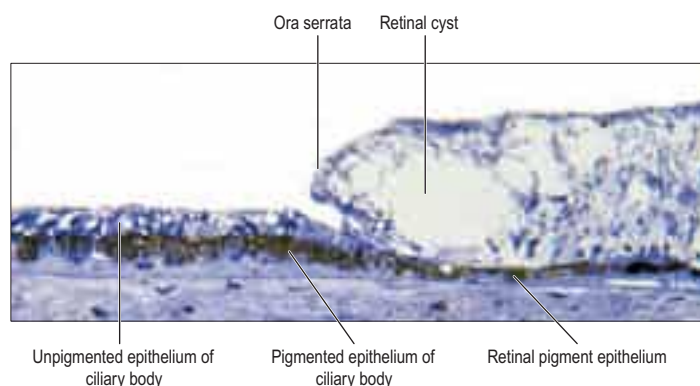


Fig. 42.24 The junction between the retina and ciliary body (ora serrata). The retinal pigment epithelium is continuous with the outer, pigmented epithelium of the ciliary body, while the neural retina abuts the inner, unpigmented epithelium of the ciliary body. The layered appearance that is apparent elsewhere in the neural retina is disrupted adjacent to the ora serrata by cystic degeneration.

Retinal pigment epithelium cells play a major role in the turnover of rod and cone photoreceptive components. Their cytoplasm contains the phagocytosed tips of rods and cones undergoing lysosomal destruction. The final products of this process are lipofuscin granules, which accumulate in these cells with age. Disturbances in this phagocytic activity of the retinal pigment epithelium can lead to retinal disease (Kevany and Palczewski 2010).

Light reaching the outer retina but missing the photoreceptors is absorbed by the retinal pigment epithelium, which, like melanin elsewhere in the eye, prevents such stray light degrading image quality. The zone of tight junctions between adjacent cells also allows the epithelium to function as an important blood–retinal barrier between the retina and the vascular system of the choroid. The retinal pigment epithelium is required for the regeneration of bleached visual pigment and may have antioxidant properties. It also secretes a variety of growth factors necessary for the integrity of the choriocapillaris endothelium and the photoreceptors, and produces a number of immunosuppressive factors. A failure of any of the diverse functions of the retinal pigment epithelium could result in compromised retinal function and eventual blindness (Strauss 2005).

Rods and cones

Rods and cones are the ‘image-forming’ photoreceptors of the outer retina and function at low (scotopic) and higher (photopic) light levels, respectively. Both are long, radially orientated structures with a similar organization, although details differ (Fig. 42.25). From the choroidal end inwards, the cells consist of outer and inner segments connected

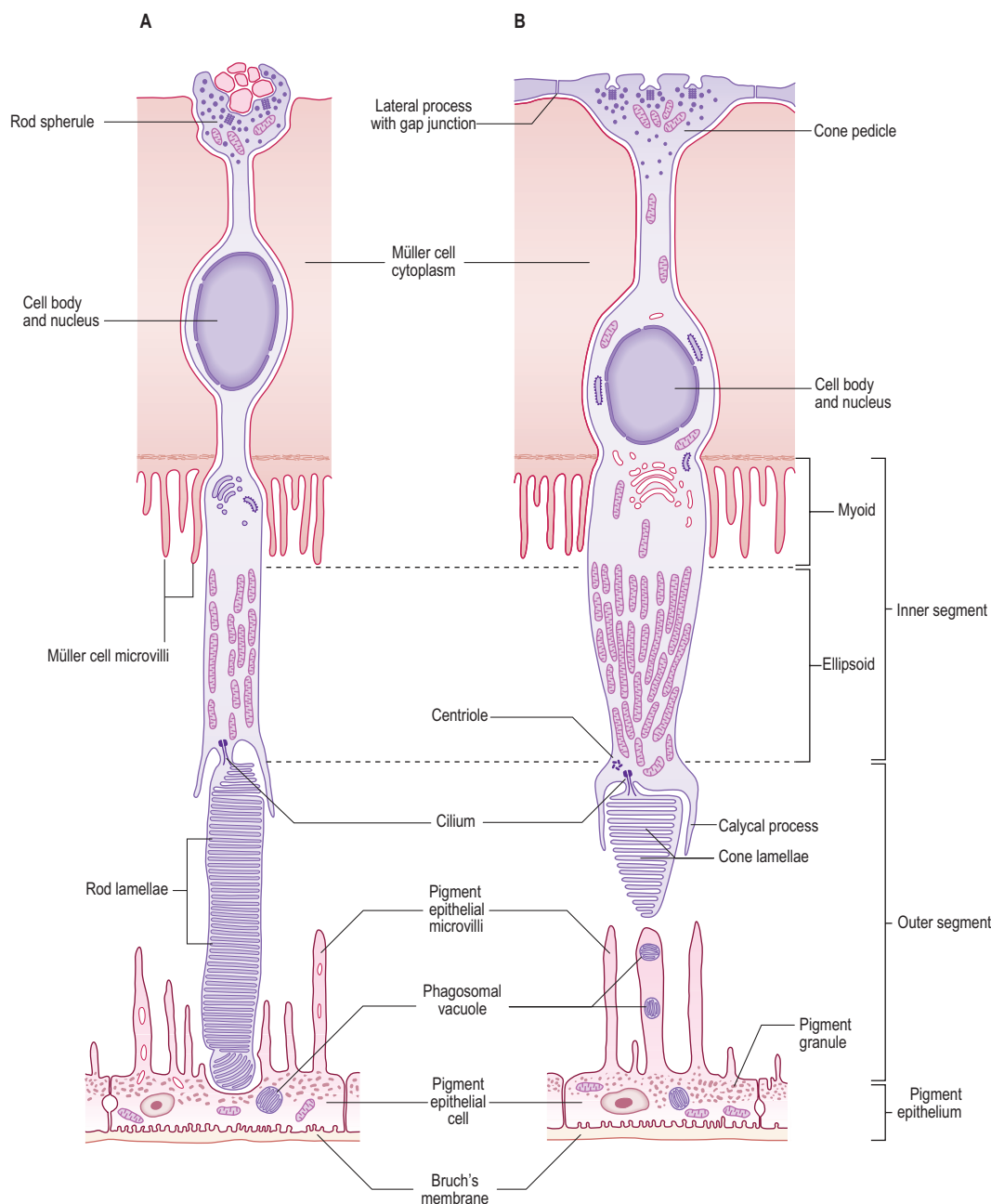


Fig. 42.25 The major features of a retinal rod cell (A) and a retinal cone cell (B). The relative size of the pigment epithelial cells has been exaggerated for illustrative purposes.

by a thin connecting cilium (together making up layer 2 of the retina), a cell body containing the nucleus, and a synaptic terminal (either a more complex pedicle for cones or a simpler rod spherule) where they make synaptic connections with adjacent bipolar and horizontal cells and with other cone or rod cells within the outer plexiform layer.

The nuclei of the rods and cones form the outer nuclear layer (layer 4). The cone nuclei are relatively large and oval, and generally form a single layer that often penetrates the external limiting membrane (layer 3). They also contain less heterochromatin and thus usually stain more lightly. Rod nuclei are round and smaller, stain more darkly, and form several layers vitreal to the cone nuclei.

The external limiting is not, in fact, a membrane at all, although it appears as such in the light microscope; rather, it is a series of zonulae adherentes between photoreceptors and the glial (Müller) cells that separate them. These junctions most likely serve to anchor the photoreceptors and prevent leakage of the interphotoreceptor matrix that surrounds the photoreceptor outer and inner segments.

Rod outer segments are cylindrical and consist of around 1000 flattened, lobulated, membranous discs. These form as deep infoldings of the plasma membrane at the base of the outer segment; they 'bud off' after formation so that the discs are not attached to the plasma membrane and are free-floating within the outer segment. Cone outer segments are generally shorter and, as their name implies, often conical (especially in the peripheral retina). Cone discs do not bud off after formation and remain as infoldings of the plasma membrane.

The inner segment of both rods and cones is divided into an outer, mitochondria-rich, ellipsoid and an inner myoid that contains endoplasmic reticulum. In most of the retina, these inner segments are much wider in cones (5–6 μm at their widest point) than rods (1.5 μm) (Fig. 42.26). In both rods and cones, proteins are manufactured within the myoid and incorporated into the newly formed discs at the base of the outer segment. In rods, as new discs are added, and old discs are pushed up the outer segment and eventually phagocytosed by the retinal pigment epithelium. Cone discs are also phagocytosed but the incorporation of new proteins within the discs is more diffuse (Nguyen-Legros and Hicks 2000). While all rods within the retina have a similar structure, the cones at the foveola are highly modified compared to those situated more peripherally and, in many ways, resemble rods with a longer outer segment and a thinner inner segment.

Light is absorbed by rhodopsins, visual pigments consisting of a protein, opsin, that spans the membrane of the outer segment discs, bound to a light-absorbing chromophore, retinal, which is an aldehyde of vitamin A₁. Such rhodopsins have a smooth, bell-shaped absorption profile with a point of maximum absorbance (λ_{max}), indicating the wavelength at which they are most sensitive. Humans possess four different opsins, resulting in four spectrally distinct visual pigments: one located within the rods (λ_{max} 498 nm) and three within different

populations of cones absorbing maximally at the short- (λ_{max} 420 nm), middle- (λ_{max} 534 nm) and longer-wave (λ_{max} 563 nm) end of the visible spectrum. The three cone classes are sometimes referred to as the blue, green and red cones but are better classed as S, M and L cones. The action of light is to isomerize the retinal, separating it from the opsin, a process which, via a G-protein coupled enzyme cascade and a second messenger system, results in the closure of cation channels in the receptor outer segment membrane, a hyperpolarization of the photoreceptor, and a consequent decrease in the release of the neurotransmitter glutamate from its synapses.

The human retina contains, on average, 4.6 million cones and 92 million rods, although there is significant inter-individual variation (Curcio et al 1990). Although cones populate the whole retina, their density is highest in the foveola, where approximately 7000 cones reach an average density of 199,000 cones/ mm^2 ; this area is entirely rod-free. Going outwards from the foveola, rod numbers rise, reaching a peak density in a horizontal elliptical ring at the eccentricity of the optic disc, before declining once more towards the periphery. Cone density is 40–45% higher in the nasal compared to the temporal retina, and slightly higher inferiorly than superiorly.

The number of S cones in all human retinæ is similar, making up less than 10% of all cones (Curcio et al 1991, Hofer et al 2005). The distribution of S cones is relatively even throughout the retina, although they are absent from the central fovea. The relative proportions of L and M cones shows a much greater degree of variation between individuals, the L:M cone ratio varying from close to unity to over 10. The distribution of L and M cones is more irregular than that of S cones, and appears random with some indication of clumping (Bowmaker et al 2003, Hofer et al 2005).

The high packing density of cones at the foveola, achieved by decreasing inner segment size, ensures maximal resolution, while the presence of more than one spectral cone type allows colour vision. S cones probably contribute little to spatial resolution because they are absent from the foveola. Rod-based vision provides high sensitivity, but with relatively low spatial discrimination and no ability to distinguish wavelengths. Although many of the functional differences between rods and cones rely on the different properties of the photoreceptors themselves, their connectivity to other retinal neurones is equally important.

Horizontal cells

Horizontal cells are inhibitory interneurons. Their dendrites and axons extend laterally within the outer plexiform layer, making synaptic contacts with cone pedicles and rod spherules, and, via gap junctions at the tips of their dendrites, with each other. Their cell bodies lie in the outer part of the inner nuclear layer (layer 6). Three morphological types of horizontal cell can be distinguished in the human retina (Kolb et al 1992). The dendrites of HI and HIII cells contact cones, and their axons terminate on rods. Both the axons and dendrites of HII cells synapse only with cones.

Bipolar cells

Bipolar cells are radially orientated neurones. Their dendrites synapse on photoreceptors, horizontal cells and interplexiform cells in the outer plexiform layer. Their somata are located in the inner nuclear layer, and axonal branches in the inner plexiform layer synapse with dendrites of ganglion cells or amacrine cells. Golgi staining has identified nine distinct types of bipolar cell in the human retina (Kolb et al 1992), eight of which contact cones exclusively, and the remaining type synapses only on rods.

Cone bipolars are of three major morphological types: midget, S (blue) cone and diffuse, according to their connectivity and size. Midget cone bipolar cells either invaginate the cone pedicle or synapse on its base (flat subtype). In the central retina, each midget bipolar cell contacts only a single cone (2–3 in the periphery), forming part of a one-to-one channel from cone to ganglion cell that mediates high spatial resolution. S cones form part of a short-wavelength mediating channel, while the larger diffuse cone bipolars are connected to up to 10 cones and are thought to signal luminosity rather than colour.

Cone bipolar cells can also be of two physiological types, according to their response to the light-induced decrease in glutamate release from the photoreceptors to which they are synaptically connected. If illumination of the photoreceptors with a point of light causes a depolarization of the connected bipolar cell, it is said to be an 'ON' bipolar, contacting the cone by 'sign-inverting' synapses with metabotropic receptors. However, if cones are connected to an 'OFF' bipolar cell via 'sign-conserving' synapses with ionotropic receptors, illumination of the photoreceptor will result in hyperpolarization of the bipolar cell. Illumination of a concentric area of surrounding photoreceptors causes the opposite response in bipolar cells to illumination within their

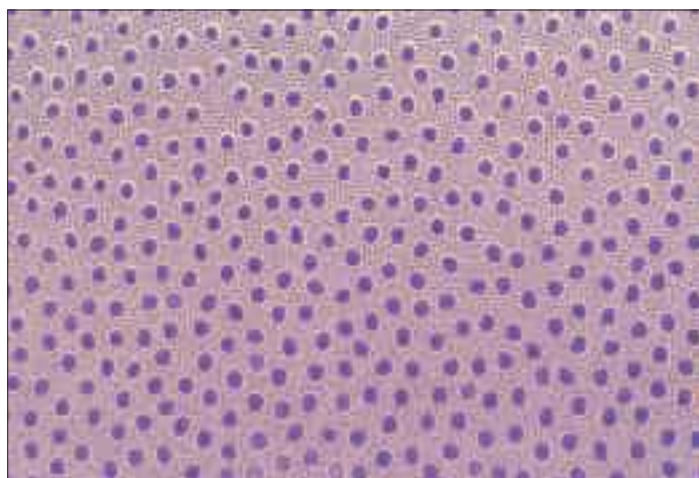


Fig. 42.26 A tangential section of a human retina in the parafovea cut at the level of the inner/outer segments, showing the 'mosaic' of both the larger cones and the more numerous, but smaller, rods. Photoreceptors at the top of the image are sectioned at a level closer to the retinal pigment epithelium than receptors lower in the figure. The reduction in size of the cones at the top of the figure is explained by the conical shape of their outer segment. If the figure were continued upwards, representing sections closer to the retinal pigment epithelium, the size of the cones would continue to decrease and the amount of surrounding white space would increase. Rod outer segment diameter, however, would change little.

dendritic field. This inhibition is mediated via horizontal cells and gives rise to the antagonistic centre-surround type receptive field that is characteristic of all levels of the visual system up to and including the occipital cortex.

The single morphological type of rod bipolar cell contacts 30–35 rods in the central retina, increasing to 40–45 rods in the periphery. Such convergence serves to increase the absolute sensitivity of the rod system. All rod bipolar cells are 'ON-centre' and do not contact ganglion cells directly, but synapse with a class of amacrine cell (AII), which then contacts cone bipolar cells.

The inner plexiform layer can be divided into two main layers: an outer layer containing the synaptic endings of 'OFF' cone bipolar cells, and an inner layer of 'ON' cone and rod bipolar cell synapses.

Amacrine cells

Most amacrine cells lack typical axons and, consequently, their dendrites make both incoming and outgoing synapses. Each neurone has a cell body either in the inner nuclear layer near its boundary with the inner plexiform layer, or on the outer aspect of the ganglion cell layer, when it is known as a displaced amacrine cell. The processes of amacrine cells make a variety of synaptic contacts in the inner plexiform layer with bipolar and ganglion cells, as well as with other amacrine cells.

The various classes of amacrine cell serve a number of important functions. All cells play an essential role in the rod pathway (see above). Other cells appear to be important modulators of photoreceptive signals, and serve to adjust or maintain relative colour and luminosity inputs under changing light conditions. They are probably also responsible for some of the complex forms of image analysis known to occur within the retina, such as directional movement detection. Up to 24 different morphological types are recognized in humans (Kolb et al 1992); coupled to their neurochemical complexity, this makes them perhaps the most diverse neural cell type in the body.

Interplexiform cells

Interplexiform cells, often regarded as a subclass of amacrine cells, generally have cell bodies in the inner nuclear layer. They are postsynaptic to cells in the inner retina, and send signals against the general direction of information flow in the retina, synapsing with bipolar, horizontal and photoreceptor cells in the outer plexiform layer. Although their function is uncertain, it is likely that, through the release of γ -aminobutyric acid (GABA) and dopamine, they adjust some aspect of retinal function such as sensitivity.

Ganglion cells

The human retina contains 0.7–1.5 million ganglion cells, the output neurones of the retina (Curcio and Allen 1990). Their dendrites synapse with processes of bipolar and amacrine cells in the inner plexiform

layer. Ganglion cell bodies, together with displaced amacrine cells, form the ganglion cell layer of the retina (layer 8). Throughout most of the retina, they form a single layer; they become progressively more numerous near the macula, where they are ranked in up to 10 rows, reaching a peak density of up to 38,000/mm² in a horizontally orientated elliptical ring 0.4–2.0 mm from the foveal centre. Their number diminishes again towards the fovea, from which they are almost totally excluded.

Up to 15 ganglion cell types have been identified in the mammalian retina based on morphology, physiology, and target area in the brain, each of them presumably functionally distinct. For example, some project to different regions of the lateral geniculate nucleus and form three parallel visual pathways involved in conscious visual perception, namely: the magnocellular and parvocellular systems and a pathway carrying the S cone signal (Wässle 2004). Midget ganglion cells (P cells) contact only single midget bipolar cells in the central retina, which, in turn, connect to single cones, giving each cone a 'private line' out of the retina and ensuring optimal acuity. The large dendritic field of parasol cells (M cells) is consistent with a role in motion detection. Parasol and midget ganglion cells together make up around 80% of human retinal ganglion cells. The remaining cells (approximately 200,000) project to the superior colliculus of the midbrain, the thalamic pulvinar, the pretectum and the accessory optic system, and contribute to various sub-conscious visual reflexes such as the pupillary and accommodation responses (see Fig. 42.12). In addition, a population of around 3000 large, intrinsically light-sensitive ganglion cells form a network composed of extensive overlapping dendrites (Dacey et al 2005). Such 'inner retinal photoreceptors' contain a retinal-based visual pigment (melanopsin; λ_{max} 479 nm), which resembles an invertebrate-type visual pigment in many of its characteristics. These light-sensitive ganglion cells are part of a pathway parallel to the rod- and cone-mediated 'image-forming' system that monitors overall levels of illumination. This 'non-imaging' pathway is the major route by which the eye influences circadian rhythms via the suprachiasmatic nucleus; it also contributes to light-evoked pupillary constriction via projections to the olivary pretectal nucleus. Although the axons of some of these photosensitive ganglion cells also project to the lateral geniculate nucleus, their wider contribution to conscious visual perception remains incompletely understood.

Ganglion cell axons, which form the nerve fibre layer on the inner surface of the retina, run parallel to the surface of the retina, and converge on the optic nerve head where they leave the eye as the optic nerve. Fibres from the medial (nasal) retina approach the disc in a simple radial pattern (Fig. 42.27), whereas axons from the lateral (temporal) retina take an arcuate route as they avoid the fovea. Axons from the macula form a papillomacular fasciculus that passes almost straight to the disc. The thickness of the nerve fibre layer increases dramatically near the optic disc as fibres from the peripheral retina traverse more central areas. Towards the edge of the disc, the other retinal layers thin,

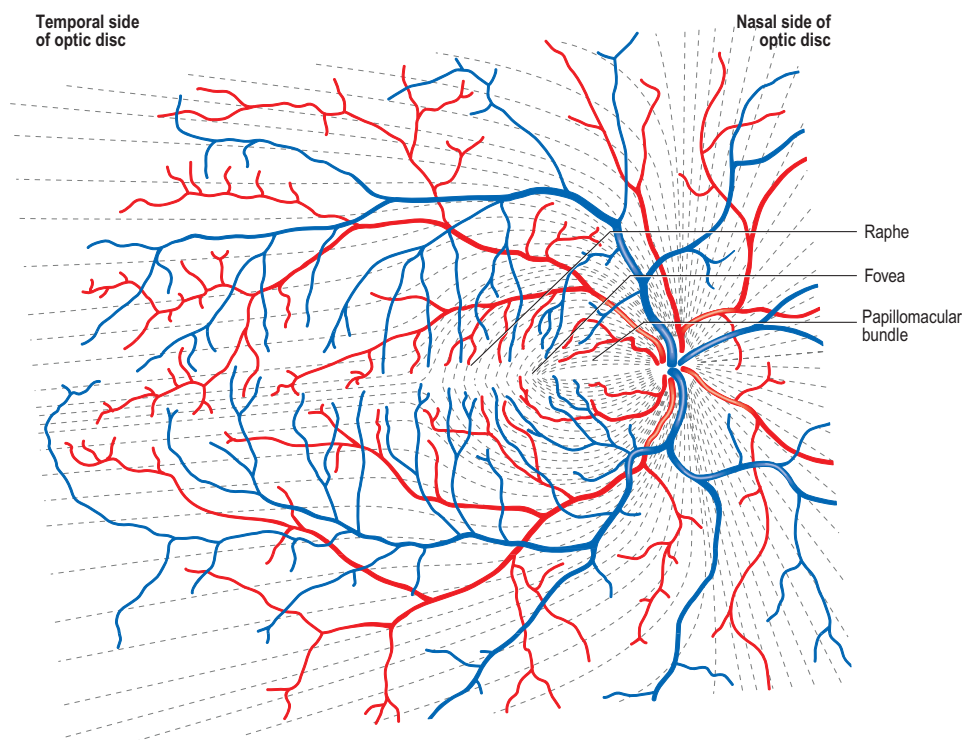


Fig. 42.27 Directions of axons (dashed lines) and blood vessels of the nerve fibre layer of the retina of the right eye. Axons pass radially on the nasal side of the optic disc, whereas fibres on the temporal side avoid crossing the fovea by arching around it. Some of the fibres from the fovea and central region pass straight to the optic disc and others arch above and below the horizontal; together, these form the papillomacular bundle. A raphe is formed by the central fibres temporal to the fovea. Venules are shown crossing in front of arteries; the reverse relationship is probably the more common pattern. All vessels issue from the disc on the right of the figure; the larger temporal branches tend to arch around the central region of the retina and do not approach the fovea. The peripheral retina and most of the nasal retina are not shown.

and, at the disc, all neural elements of the retina other than ganglion cell axons are excluded.

Axons of ganglion cells are surrounded by the processes of radial glial cells and retinal astrocytes, and are almost always unmyelinated within the retina, which is an optical advantage because myelin is refractile. Although a few small myelinated axons may occur, myelination does not generally start until axons enter the optic disc to become the optic nerve.

Retinal glial cells

There are three types of retinal glial cells: radial Müller cells, astrocytes and microglia. Müller cells form the predominant glial element of the retina; retinal astrocytes are largely confined to the ganglion cell and nerve fibre layers; and microglial cells are scattered throughout the neural part of the retina in small numbers.

Müller cells span almost the entire thickness of the neural retina, ensheathing and separating the various neural cells except at synaptic sites. They constitute much of the total retinal volume, and almost totally fill the extracellular space between neural elements. Their nuclei lie within the inner nuclear layer, and from this region each cell body extends a single thick fibre that runs radially outwards, giving off complex lateral lamellae that branch among the processes of the outer plexiform layer. Apically, each central process terminates at the external limiting membrane, from which microvilli project for a short distance into the space between the rod and cone inner segments (fibre baskets) (see Fig. 42.25). On the inner surface of the retina, the main Müller cell process expands into a terminal foot plate that contacts those of neighbouring glial cells and forms part of the internal limiting membrane (see below).

The functions of Müller cells are numerous (Reichenbach and Briggmann 2013). Like astrocytes, Müller cells contact blood vessels, especially capillaries of the inner nuclear layer, and their basal laminae fuse with those of perivascular cells or vascular endothelia, contributing to the formation of the blood–retinal barrier. They also maintain the stability of the retinal extracellular environment by, for example, regulation of K^+ levels, uptake of neurotransmitter, removal of debris, storage of glycogen, providing neuroprotective support to the photoreceptors and mechanical support to the whole neural retina. Recently, it has been shown that they are also involved in the regeneration of cone visual pigments, that some are a source of stem cells and that they may even act as light guides, conveying light from the inner retina to the photoreceptors and overcoming some of the optical disadvantages of an ‘inverted’ retina.

The cell bodies of retinal astrocytes lie within the nerve fibre layer and their processes branch to form sheaths around ganglion cell axons. The close association between astrocytes and blood vessels in the inner retina suggests that they contribute to the blood–retinal barrier. Retinal microglia are scattered mostly within the inner plexiform layer. Their radiating branched processes spread mainly parallel to the retinal plane, giving them a star-like appearance when viewed microscopically from the surface of the retina. They can act as phagocytes, and their number increases in the injured retina.

The inner border of the retina is formed by the internal limiting membrane (layer 10), which consists of collagen fibres and proteoglycans from the vitreous, a basement membrane (which is continuous with the basal lamina of the ciliary epithelium), and the plasma membrane of expanded Müller cell terminal foot plates. It is 0.5–2 μm thick in the posterior retina and thickens with age. The internal limiting membrane is involved in fluid exchange between the vitreous and the retina, and, perhaps through the latter, with the choroid. It also has various other functions, including anchorage of retinal glial cells, and inhibition of cell migration into the vitreous body.

Modifications of the central retina

The central retina, clinically referred to as the macula, is composed of four concentric areas, which, starting with the innermost, are: the foveola (0.35 mm diameter, equivalent of an angular subtense at the nodal point of around 1.25°), the fovea (1.5 mm, 5.2°), the parafovea (2.5 mm, 8.6°) and the vaguely defined perifovea (5–6 mm, 20°). The foveola, which contains no rods or S cones, is centred about 3 mm temporal and 1 mm inferior to the optic disc (see Fig. 42.20A). In the foveola and surrounding fovea, all the inner layers of the neural retina beyond the outer nuclear layer have been displaced peripherally, resulting in a retinal thickness around half of that elsewhere in the retina (see Fig. 42.23; Fig. 42.28). This foveal pit is created by the cone ‘axons’, known here as Henle fibres, running almost parallel to the retinal surface before connecting to postreceptoral retinal neurones outside the

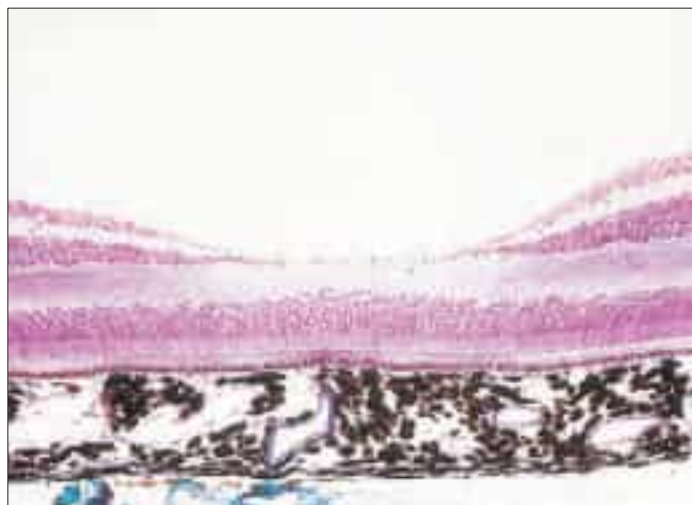


Fig. 42.28 A section through the fovea centralis. (With permission from Young B, Heath JW 2000 Wheater's Functional Histology. Edinburgh: Churchill Livingstone.)

fovea. The Henle fibres contain two xanthophyll carotenoid pigments (lutein and zeaxanthin), which create an elliptical yellowish area (approximately 2 mm horizontally and 1 mm vertically): the macula lutea. Macular pigment density varies by more than an order of magnitude between individuals, is influenced by several environmental factors, including diet, and is negligible in the central foveola. Low levels of macular pigment are likely to be associated with retinal pathologies such as age-related macular degeneration (Beatty et al 2008).

The absence of the inner retinal layers, including blood vessels (see Fig. 42.20A), reduces light scatter, which, along with the increased packing density of cones in the foveola and their lack of convergence with ganglion cells, ensures that visual resolution is highest in this part of the retina. Acuity may be further enhanced by the macular pigment, which, apart from having antioxidant properties and removing potentially harmful short-wave radiation, will absorb those wavelengths most prone to chromatic aberration and Rayleigh scatter.

VASCULAR SUPPLY

The retina has a dual arterial supply and both parts are necessary to maintain retinal function. The outer five layers of the retina are avascular and rely on an indirect supply from the choroidal capillaries. The inner retina receives a direct blood supply through capillaries connected to branches of the central retinal artery and vein. Only the inner retinal circulation is described here.

The central retinal artery enters the optic nerve as a branch of the ophthalmic artery 6.4–15.2 mm behind the eyeball (Kocabiyik et al 2005), and travels within the optic nerve to its head, where it passes through the lamina cribrosa. At this level, the central artery divides into equal superior and inferior branches, which, after a few millimetres, divide into superior and inferior nasal, and superior and inferior temporal, branches, each supplying a ‘quadrant’ of the retina (see Fig. 42.27). Although similar retinal veins unite to form the central retinal vein, the courses of the arteries and veins do not correspond exactly. These vessels mainly run within the nerve fibre and ganglion cell layers of the retina, accounting for their clarity when seen through an ophthalmoscope (see Fig. 42.20A). Arteries often cross veins, usually lying superficial to them; in severe hypertension, the arteries may press on the veins and cause visible dilations distal to these crossings. The vitreal location of arteries, their lighter, bright red colouration and smaller diameter in comparison to veins allow the two vessel types to be distinguished ophthalmoscopically.

From the four major arteries within the inner retina, dichotomous branches run from the posterior pole to the periphery, supplying the whole retina (Zhang 1994). Arteries and veins ramify in the nerve fibre layer, near the internal limiting membrane, and arterioles pass deeper into the retina to supply capillary beds. Venules return from these beds to larger superficial veins that converge towards the disc to form the central retinal vein.

Retinal capillary networks can occur in three different layers, the number of layers depending on location (Zhang 1994). Radial peripapillary capillaries are the most superficial of the capillary networks

and lie within the inner nerve fibre layer. A layer of inner capillaries runs within the nerve fibre and ganglion cell layers, while an outer capillary layer is located in the inner plexiform and inner nuclear layers (Fig. 42.29). Approaching the fovea, capillaries are restricted to two layers, and terminal capillaries eventually join to form a single-layered macular capillary ring, producing a capillary-free zone 450–500 μm in diameter at the fovea. This avascular region is clearly visible in a fluorescein angiogram (Fig. 42.20B). Capillaries become less numerous in the peripheral retina and are absent from a zone approximately 1.5 mm wide adjoining the ora serrata.

The territories of the arteries that supply a particular quadrant do not overlap, nor do the branches within a quadrant anastomose with each other; consequently, a blockage in a retinal artery causes loss of vision in the corresponding part of the visual field. The only exception to this end-arterial pattern is in the vicinity of the optic disc. Here, the posterior ciliary arteries enter the eye near the disc (Fig. 42.30), and their rami not only supply the adjacent choroid, but also form an anastomotic circle in the sclera around the head of the optic nerve. Branches from this ring join the pial arteries of the nerve, and small cilioretinal arteries from any arteries in this region may enter the eye and contribute to the retinal vasculature, possibly resulting in the preservation of visual function following central retinal artery occlusion. Similarly, small retinociliary veins may sometimes also be present.

The structure of retinal blood vessels resembles that of vessels elsewhere, except that the internal elastic lamina is absent from the arteries, and muscle cells may appear in their adventitia. Capillaries are non-fenestrated and endothelial cells are joined by complex tight junctions, fulfilling the requirements of a functional blood–retinal barrier. Within the optic nerve, the central artery is innervated by both branches of the autonomic nervous system; this innervation does not extend to the vessels in the retina. The cholinergic parasympathetic supply is derived mainly from the pterygopalatine ganglion and is vasodilatory (Ch. 32).

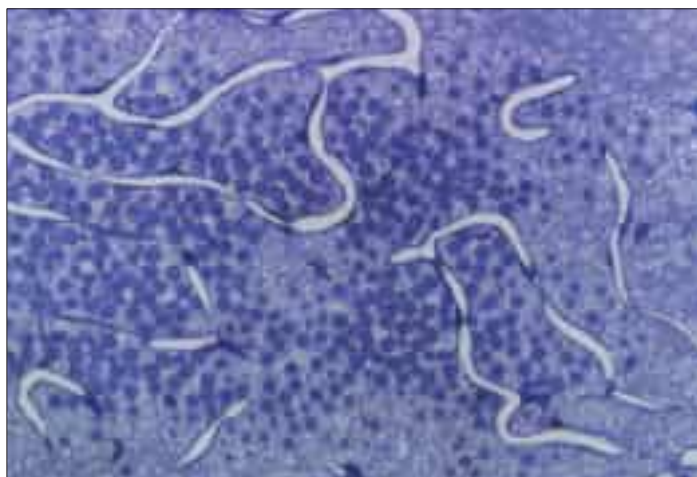


Fig. 42.29 A tangential section of the retina at the level of the inner nuclear layer, highlighting the dense network of capillaries.

The adrenergic postganglionic sympathetic supply originates in the superior cervical ganglion and travels via a plexus around the internal carotid and ophthalmic arteries; it is unclear whether it elicits vasoconstriction or dilation (Bergua et al 2013).

RETINOPETAL INNERVATION OF THE RETINA

The human retina receives input from the brain via retinopetal axons within the optic nerve. Although there are only a small number of these (usually less than 10), they branch extensively within the retina. They emerge at the optic disc, course through the nerve fibre layer, and give off orthogonal branches that ramify in the inner plexiform layer (Repérant et al 2006). One set of such axons arises from perikarya in the posterior hypothalamus and uses histamine as a neurotransmitter, while other, serotonergic, fibres arise from cell bodies in the dorsal raphe. These neurones are not specialized for vision; they project to many other targets in the central nervous system besides the retina, forming components of an ascending arousal system (Gastinger et al 2006). Histamine is released during the day in the inner plexiform layer, but it activates receptors located on cone pedicles, horizontal cell perikarya and ON bipolar cell dendrites via volume transmission. Histamine decreases the absolute sensitivity of the retina to light and, possibly, modulates retinal blood flow (Vila et al 2012).

OPTIC NERVE HEAD

The axons of more than a million retinal ganglion cells converge at the optic nerve head and leave the eye by penetrating the sclera to form the optic nerve. The optic nerve head represents that part of the optic nerve lying within the bulb of the eye. Since all retinal neural elements, apart from ganglion cell axons, are absent from this region, it is insensitive to light and forms the 'blind spot'.

Histologically, the optic nerve head can be divided into three zones (Fig. 42.31): the prelaminar (the anterior part terminating at the vitreous), laminar (formed by the lamina cribrosa) and postlaminar (continuous with the retrobulbar optic nerve). The surface view of the optic nerve head, usually seen with an ophthalmoscope, is referred to as the optic disc (Fig. 42.20).

Prelaminar zone

The inner surface of the optic nerve head is covered by an astroglial membrane (of Elschnig) that is continuous with the internal limiting membrane of the retina. At the centre of the disc, the layer of astrocytes thickens into a central meniscus (of Kuhnt). Retinal ganglion cells turn into the optic nerve head accompanied by astrocytes, which gradually increase in number posteriorly, eventually forming a sieve-like structure, the glial lamina cribrosa, through which the nerve fibres pass as separate fasciculi. At the perimeter of the optic nerve head, a collar of astrocytes several cells thick (the intermediary tissue of Kuhnt) separates the optic nerve from the terminating outer layers of the retina. This layer continues posteriorly and forms a barrier between the optic nerve head and the choroid (the border tissue of Jacoby).

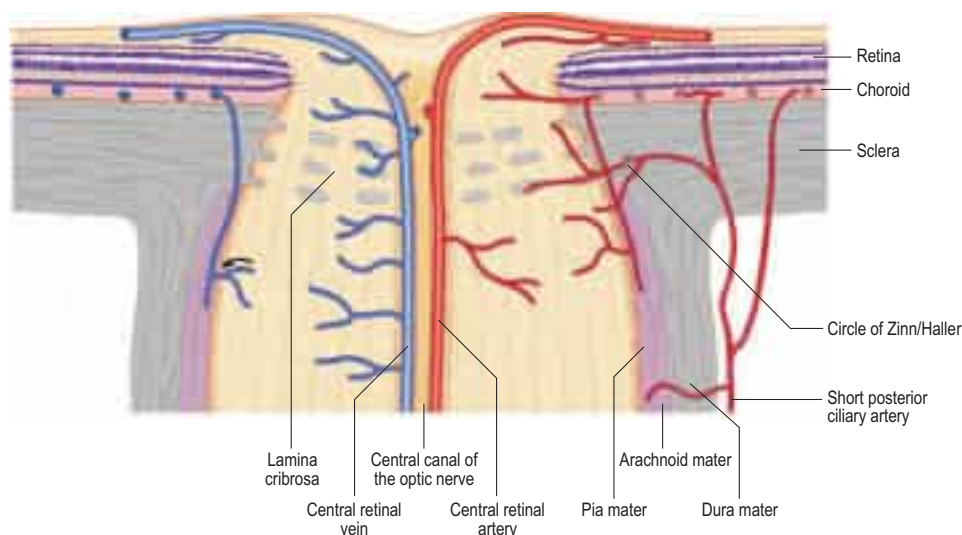


Fig. 42.30 Vessels of the optic nerve head. The dotted areas represent the principal glial membranes. Arteries are represented on the right and veins on the left.

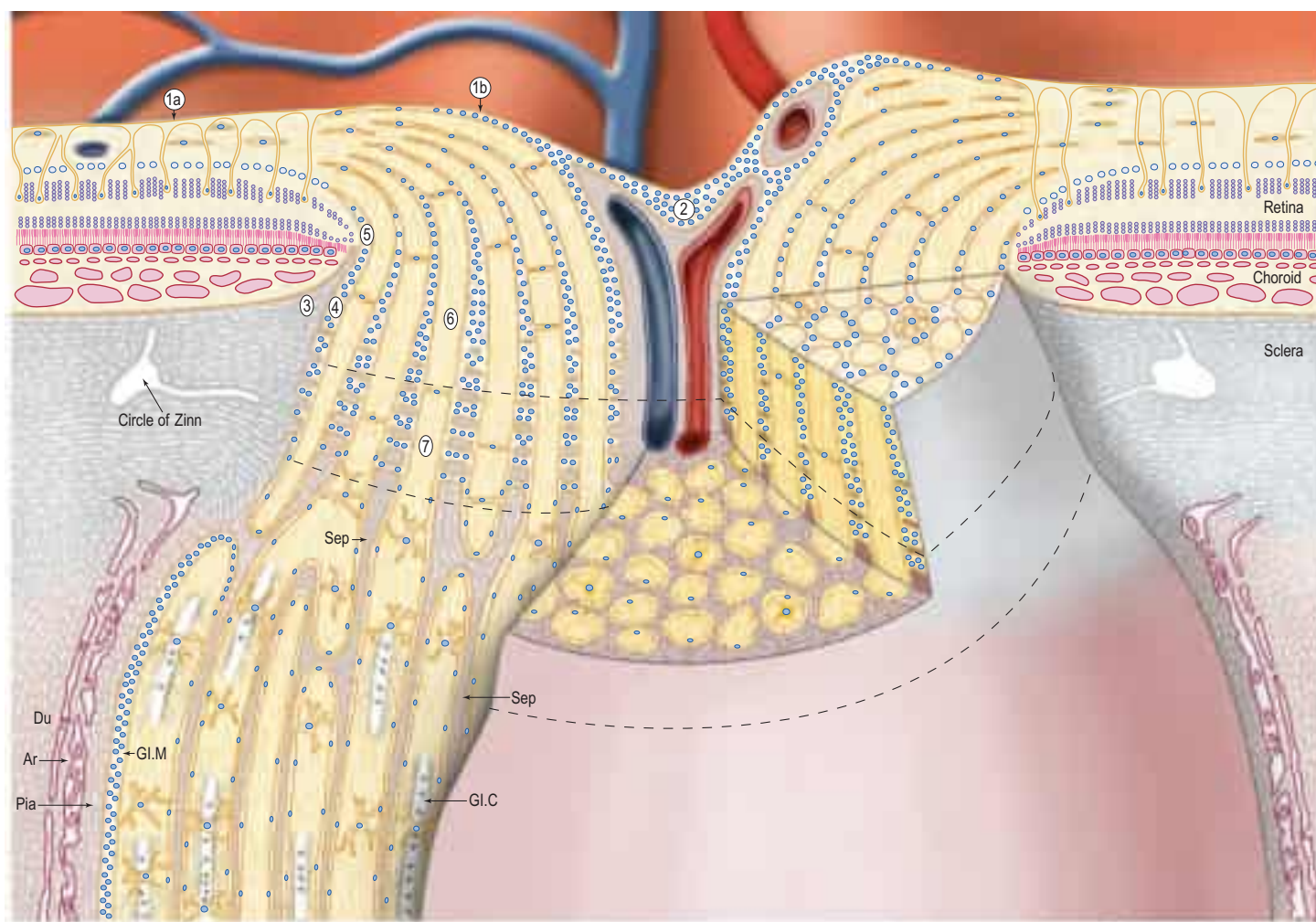


Fig. 42.31 The optic nerve head, showing the distribution of collagenous tissue (grey) and neuroglial nuclei (solid blue circles). Key: 1a, retinal internal limiting membrane; 1b, inner limiting membrane of Elschnig; 2, central meniscus of Kuhnt; 3, spur of collagenous tissue separating the anterior lamina cribrosa (6) from the choroid; 4, border tissue of Jacoby; 5, intermediary tissue of Kuhnt; 7, posterior lamina cribrosa. Abbreviations: Ar, arachnoid mater; Du, dura mater; Gl.C, astrocytes and oligodendrocytes among the fibres in their fascicles; Gl.M, astroglial membrane; Pia, pia mater; Sep, connective tissue septa from pia mater. The dotted lines represent the borders of the lamina cribrosa. (With permission from Anderson DR, Hoyt W 1969 Ultrastructure of intraorbital portion of human and monkey optic nerve. *Arch Ophthalmol* 82:506–30.)

Laminar zone

The lamina cribrosa is composed of discrete trabeculae of collagenous and elastic connective tissue, which extend from the sclera to form a meshwork through which the optic nerve fascicles and central retinal vessels pass. Each trabecula has a lining of astrocytes that are continuous with those of the glial lamina cribrosa.

Postlaminar zone

The optic nerve thickens in the postlaminar zone as its axons become myelinated. The reflected sclera, and the dura mater with which it is continuous, invest the nerve together with the other two meningeal sheaths, the arachnoid and pia mater. Fine fibrous septa penetrate the optic nerve from the pia mater, dividing it into 300–400 fascicles, giving pial blood vessels access to the nerve.

Optic disc

As it is visible by ophthalmoscopy (see Fig. 42.20A), the disc is a region of great clinical importance. Oedema of the disc (papilloedema) may be the first sign of raised intracranial pressure, which is transmitted into the subarachnoid space around the optic nerve. The disc is also sensitive to the raised intraocular pressure that occurs in glaucoma and shows characteristic structural changes due to retinal ganglion cell loss.

The optic disc is superomedial to the posterior pole of the eye, and so lies away from the visual axis. It is round or oval, and usually approximately 1.6 mm in transverse diameter and 1.8 mm in vertical diameter; its appearance is very variable. In light-skinned subjects, the

general retinal hue is a bright terracotta-red, with which the pale pink of the disc contrasts sharply; its central part is usually even paler and may be light grey. These differences are due in part to the degree of vascularization of the two regions, which is much less at the optic disc, and also to the total absence of choroidal or retinal pigment cells. In subjects with strongly melanized skins, both retina and disc are darker. The optic disc rarely projects sufficiently to justify the term papilla, although it is usually a little elevated on its lateral side, where the papillomacular nerve fibres turn into the optic nerve (see Fig. 42.27). There is usually a slight depression where the retinal vessels traverse its centre.

Vascular supply

The blood supply to the three regions of the optic nerve head differs (see Fig. 42.30). The prelaminar region is supplied mainly by branches of the central retinal artery. Branches from the short posterior ciliary arteries form an often incomplete circle within the sclera around the optic nerve head (circle of Zinn/Haller); centripetal branches from this structure supply the laminar region of the optic nerve head. The short posterior ciliary arteries may also give off centripetal branches directly to supply the lamina, and branches that pass anteriorly to augment the prelaminar blood supply. In the postlaminar region, arteries from the prepapillary choroid and circle of Zinn pass retrogradely as pial vessels, providing centripetal branches that supply the optic nerve. More posteriorly, the optic nerve receives pial arterioles directly from the posterior ciliary arteries. The central retinal artery may also contribute some centrifugal branches in this region.

The central retinal vein drains the optic nerve head at all levels; other drainage pathways are minor.

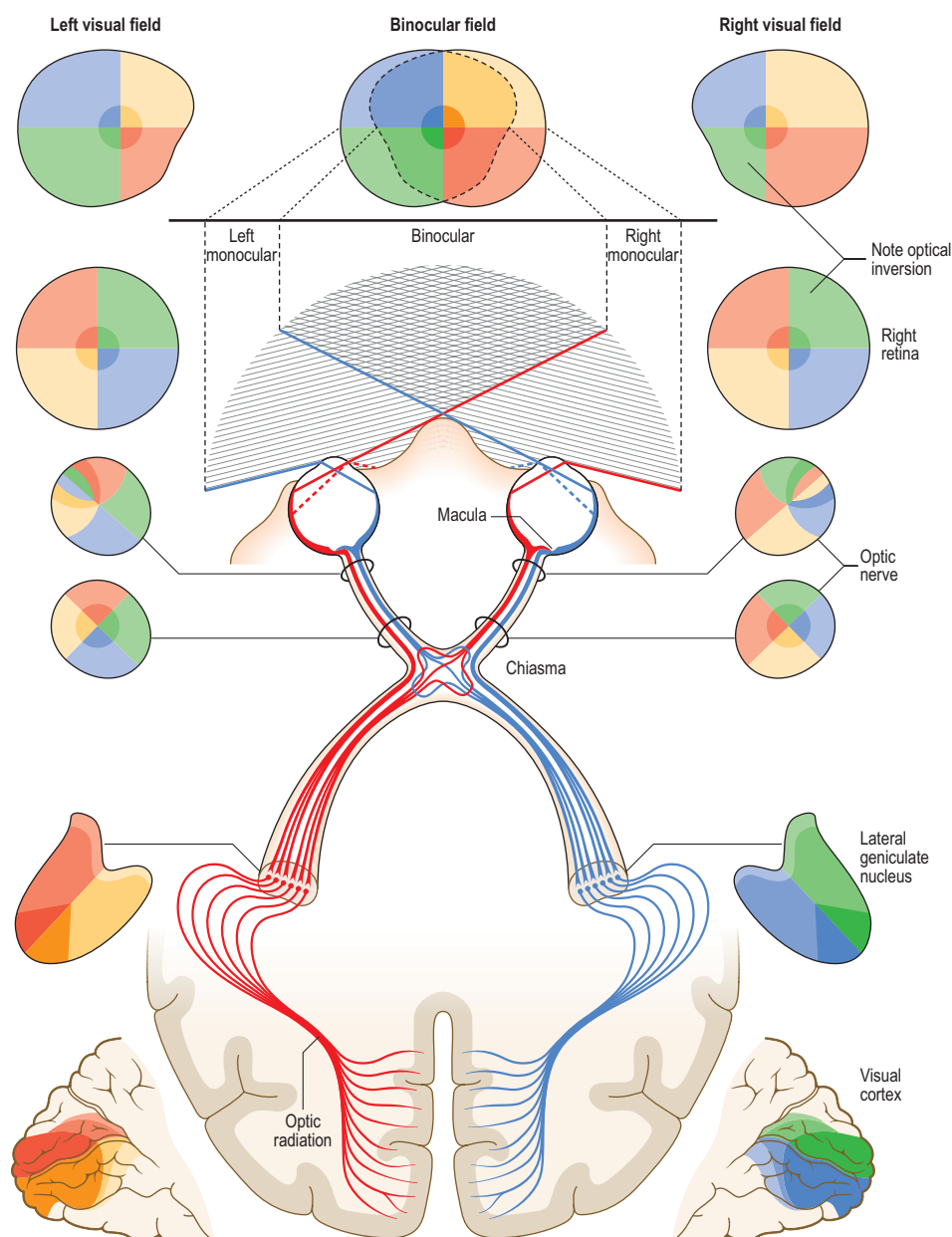


Fig. 42.32 A simplification of the visual pathway, showing the spatial arrangement of neurones and their fibres in relation to the quadrants of the retinae and visual fields. The proportions at various levels are not exactly to scale. In particular, the macula is exaggerated in size in the visual fields and retinae. In each quadrant of the visual field, and in the parts of the visual pathway subserving it, two shades of each respective colour are used; the paler shade denotes the peripheral field and the darker shade denotes the macular part of the quadrant. From the lateral geniculate nucleus onwards, these two shades are both made more saturated to denote intermixture of neurones from both retinae, the palest shade being reserved for parts of the visual pathway concerned with monocular vision.

VISUAL PATHWAY

The visual pathway includes the interneurons of the retina, retinal ganglion cells whose axons project via the optic nerve, chiasma, and optic tract to the lateral geniculate nucleus and neurones within the lateral geniculate nucleus that project via the optic radiation to the primary visual cortex (Fig. 42.32). It is important to remember that visual space is optically inverted by the crystalline lens when relating the spatial location of neurones within the visual pathway to corresponding visual field locations.

Retinal ganglion cell axons, on entering the optic nerve, initially maintain their relative retinal positions, with axons from the fovea forming a lateral wedge. Such retinotopic mapping is largely maintained within the optic nerve, although nearer the chiasma the foveal axons take a position in the centre of the optic nerve while temporal fibres occupy their previous lateral location.

At the chiasma, a substantial rearrangement of axons occurs. Most axons arising from the nasal half of a line bisecting the fovea within each retina cross in the chiasma to enter the contralateral optic tract. Fibres from the temporal hemi-retinas do not generally cross in the chiasma. Classically, the axons within the optic tract were thought to maintain their topographic order and each tract was assumed to be a single representation of the contralateral hemifield. However, it is now clear that axons are mainly organized in functional groupings, larger superficial axons representing the magnocellular pathway and deeper axons originating from midget ganglion cells and forming the parvocellular pathway. This arrangement is chronotopic, the deeper axons

developing earlier during axogenesis than the more superficial ones (Reese 1993).

The lateral geniculate nucleus contains cells arranged in six laminae (see Fig. 23.6). Each layer receives input from either crossed or uncrossed projections from the retina. The contralateral nasal retina projects to laminae 1, 4 and 6, whereas the ipsilateral temporal retina projects to layers 2, 3 and 5. Layers 1 and 2 contain magnocellular cells; the remaining layers are parvocellular. Unlike in the optic tract, there is a point-to-point retinotopic arrangement between corresponding points in each hemi-retina so that the contralateral visual field is mapped within each lateral geniculate nucleus.

Axons from the lateral geniculate nucleus run in the retrolenticular part of the internal capsule and form the optic radiation. This curves dorsomedially to the primary visual cortex, located around and within the depths of the calcarine sulcus in the occipital lobe (also known as the striate cortex, Brodmann area 17, or V1 (see Fig. 25.19)). The visual cortex also has a strict retinotopic organization. Fibres representing the lower half of the visual field sweep superiorly to reach the visual cortex above the calcarine sulcus, while those representing the upper half of the visual field curve inferiorly into the temporal lobe (Meyer's loop) before reaching the visual cortex below the calcarine sulcus. The periphery of the retina is represented anteriorly within the visual cortex, and the macula is represented towards the posterior pole, occupying a disproportionately large area that reflects the high number of foveal retinal ganglion cells that subserve the enhanced acuity of this region.

The primary visual cortex is connected to prestriate and other cortical regions where further processing of visual stimuli occurs.

Almost all of the retinal ganglion cell axons (90%) terminate on neurones in the lateral geniculate nucleus. Extrageniculate axons (10%) leave the optic tract before the lateral geniculate nucleus; they may leave the optic chiasma dorsally and project to the suprachiasmatic nucleus of the hypothalamus, while others branch off the optic tract at the superior brachium and project to the superior colliculus, pretectal areas and inferior pulvinar.

VISUAL FIELD DEFECTS

The basis for clinical assessment of damage to the visual pathway is an understanding of the retinotopic projections within the pathway. Moreover, plotting visual field loss frequently reveals the approximate location of the causative lesion and sometimes its nature. Since retinal lesions can be visualized with an ophthalmoscope, field testing might appear to be redundant for such defects, but visual field measurement is still helpful in assessing the extent of the damage and may be the key factor in confirming a diagnosis. Field defects in glaucoma, for example, that occur as a consequence of damage to the nerve fibre bundles at the optic nerve head, may be detectable ophthalmoscopically, but confirmation of the diagnosis frequently depends on field assessment. Early defects consist of one or more areas of paracentral focal field loss, progressing to arcuate scotomas. The shape of the defect corresponds to the anatomical arrangement of ganglion cell axons.

As far as the location of lesions central to the retina is concerned, deficits in the vision of one eye are usually attributable to optic nerve lesions. Lesions of the optic chiasma, involving crossing nerve fibres, produce a bilateral field loss, as exemplified by a pituitary adenoma. The tumour expands upwards from the pituitary fossa, compressing the inferior midline of the chiasma, and eventually produces bitemporal hemianopia, starting with an early loss in the upper temporal quadrants (bitemporal quadrantanopia).

Since the optic tract contains contralateral nasal and ipsilateral temporal retinal projections, damage to it will cause a homonymous contralateral visual field loss. Although complete disruption of the tract results in contralateral hemianopia, the incomplete spatial segregation of axons within the tract, described above, makes the field losses following smaller lesions harder to interpret. They do, however, show substantial incongruity (dissimilar defects in the fields of the two eyes) and often specific functional deficits, consistent with the partial segregation of functionally distinct axons from the two half-retinas (Reese 1993). Incongruity is most marked in defects of the optic tract, less obvious in optic radiation defects, and usually absent in cortically induced field defects, thus providing an additional clue in assessing location of the cause.

Lesions of the optic radiations are usually unilateral, and commonly vascular in origin. Field defects therefore develop abruptly, in contrast to the slow progression of defects associated with tumours, and the resulting hemifield loss follows the general rule that visual field defects central to the chiasma are on the opposite side to the lesion. Little or no incongruity is seen in visual cortical lesions, but they commonly display the phenomenon of macular sparing, the central 5–10° field being retained in an otherwise hemianopic defect.



Bonus e-book image

Fig. 42.20B Fluorescein angiogram showing the macular region of a right eye. The main macular vessels are approaching from the right. The subject was an elderly person with considerable macular pigmentation, which masks fluorescence from the choroidal circulation.

KEY REFERENCES

- Arora KS, Jefferys JL, Maul EA et al 2012 The choroid is thicker in angle closure than in open angle and control eyes. *Inv Ophthalmol Vis Sci* 53:7813–18.
An investigation of some of the factors affecting choroidal thickness measured in vivo.
- Boote C, Dennis S, Huang YF et al 2005 Lamellar orientation in human cornea in relation to mechanical properties. *J Struct Biol* 149:1–6.
The quantification by X-ray scattering of the arrangement of collagen fibres in the human corneal stroma.
- Charman WN 2008 The eye in focus: accommodation and presbyopia. *Clin Exp Optom* 91:207–25.
A summary of the various mechanisms of accommodation and the reasons for the decline of accommodative amplitude with age.
- Curcio CA, Sloan KR, Kalina RE et al 1990 Human photoreceptor topography. *J Comp Neurol* 292:497–523.
The first description of the distribution of rods and cones in the human retina using modern methods.
- Dacey DM, Liao H-S, Peterson BB et al 2005 Melanopsin-expressing ganglion cells in primate retina signal colour and irradiance and project to the LGN. *Nature* 433:749–54.
Although directly light-sensitive melanopsin-containing retinal ganglion cells were first described in rodents, this is the first description of their occurrence in primates.
- Kolb H, Linberg KA, Fisher SK 1992 Neurons of the human retina – a Golgi study. *J Comp Neurol* 318:147–87.
The most comprehensive description of the morphology of neural cell types in the human retina.
- Reichenbach A, Bringmann A 2013 New functions of Müller cells. *Glia* 61:651–78.
A review of the many functions of Müller cells, some of them only recently proposed.
- Strauss O 2005 The retinal pigment epithelium in visual function. *Phys Rev* 85:845–81.
A review of the many functions of the retinal pigment epithelium and their role in various disease processes.
- Wallman J, Winawer J 2004 Homeostasis of eye growth and the question of myopia. *Neuron* 43:447–68.
A review of the mechanisms of emmetropization and the causes of myopia.
- Wang D, Qi M, He M et al 2012 Ethnic differences of anterior chamber area and volume and its association with angle width. *Inv Ophthalmol Vis Sci* 53:3139–44.
An optical coherence tomography (OCT) investigation showing differences in the morphology of the anterior chamber between Caucasians and people with a Chinese heritage.

REFERENCES

- Alm A, Nilsson SFE 2009 Uveoscleral outflow – a review. *Exp Eye Res* 88:760–8.
- Arora KS, Jefferys JL, Maul EA et al 2012 The choroid is thicker in angle closure than in open angle and control eyes. *Inv Ophthalmol Vis Sci* 53:7813–18.
An investigation of some of the factors affecting choroidal thickness measured in vivo.
- Beatty S, van Kuijk FJGM, Chakravarthy U 2008 Macular pigment and age-related macular degeneration: longitudinal data and better techniques of measurement are needed. *Invest Ophthalmol Vis Sci* 49:843–5.
- Bergua A, Kapsreiter M, Neuhuber WL et al 2013 Innervation pattern of the preocular human central retinal artery. *Exp Eye Res* 110:142–7.
- Boote C, Dennis S, Huang YF et al 2005 Lamellar orientation in human cornea in relation to mechanical properties. *J Struct Biol* 149:1–6.
The quantification by X-ray scattering of the arrangement of collagen fibres in the human corneal stroma.
- Bowmaker JK, Parry JWL, Mollon JD 2003 The arrangement of L and M cones in human and a primate retina. In: Mollon JD, Pokorny J, Knoblauch K (eds) *Normal and Defective Colour Vision*. Oxford: Oxford University Press.
- Charman WN 2008 The eye in focus: accommodation and presbyopia. *Clin Exp Optom* 91:207–25.
A summary of the various mechanisms of accommodation and the reasons for the decline of accommodative amplitude with age.
- Curcio CA, Allen KA 1990 Topography of ganglion cells in human retina. *J Comp Neurol* 300:5–25.
- Curcio CA, Allen KA, Sloan KR et al 1991 Distribution and morphology of human cone photoreceptors stained with anti-blue opsin. *J Comp Neurol* 312:610–24.
- Curcio CA, Sloan KR, Kalina RE et al 1990 Human photoreceptor topography. *J Comp Neurol* 292:497–523.
The first description of the distribution of rods and cones in the human retina using modern methods.
- Dacey DM, Liao H-S, Peterson BB et al 2005 Melanopsin-expressing ganglion cells in primate retina signal colour and irradiance and project to the LGN. *Nature* 433:749–54.
Although directly light-sensitive melanopsin-containing retinal ganglion cells were first described in rodents, this is the first description of their occurrence in primates.
- Fautsch MP, Johnson DH 2006 Aqueous humor outflow: What do we know? Where will it lead us? *Inv Ophthalmol Vis Sci* 47:4181–7.
- Flitcroft DI 2013 Is myopia a failure of homeostasis? *Exp Eye Res* 114:16–24.
- Friling R, Weinberger D, Kremer I et al 2004 Keratometry measurements in preterm and full term newborn infants. *Br J Ophthalmol* 88:8–10.
- Gastinger MJ, Tian N, Horvath T et al 2006 Retinopetal axons in mammals: emphasis on histamine and serotonin. *Curr Eye Res* 31:655–67.
- Goel M, Picciani RG, Lee RK et al 2010 Aqueous humor dynamics: a review. *Open Ophthalmol J* 4:52–9.
- Hofer H, Carroll J, Neitz J et al 2005 Organization of the human trichromatic cone mosaic. *J Neurosci* 25:9669–79.
- Kevany BM, Palczewski K 2010 Phagocytosis of retinal rod and cone photoreceptors. *Physiology* 25:8–15.
- Kocabiyik N, Yalcin B, Ozan H 2005 The morphometric analysis of the central retinal artery. *Ophthalmol Opt* 25:375–8.
- Kolb H, Linberg KA, Fisher SK 1992 Neurons of the human retina – a Golgi study. *J Comp Neurol* 318:147–87.
The most comprehensive description of the morphology of neural cell types in the human retina.
- Müller LJ, Marfurt CF, Kruse F et al 2003 Corneal nerves: structure, contents and function. *Exp Eye Res* 76:521–42.
- Mutti DO, Zadnik K, Fusaro RE et al 1998 Optical and structural development of the crystalline lens in childhood. *Invest Ophthalmol Vis Sci* 39:120–33.
- Nguyen-Legros J, Hicks D 2000 Renewal of photoreceptor outer segments and their phagocytosis by the retinal pigment epithelium. *Int Rev Cytol* 196:245–313.
- Park KA, Oh SY 2013 Choroidal thickness in healthy children. *Retina* 33:1971–6.
- Reese BE 1993 Clinical implications of the fibre order in the optic pathway of primates. *Neurol Res* 15:83–6.
- Reichenbach A, Bringmann A 2013 New functions of Müller cells. *Glia* 61:651–78.
A review of the many functions of Müller cells, some of them only recently proposed.
- Repérant J, Ward R, Miceli D et al 2006 The centrifugal visual system of vertebrates: a comparative analysis of its functional anatomical organization. *Brain Res Rev* 52:1–57.
- Rose KA, Morgan IG, Smith W 2008 Myopia, lifestyle, and schooling in students of Chinese ethnicity in Singapore and Sydney. *Arch Ophthalmol* 126:527–30.
- Strauss O 2005 The retinal pigment epithelium in visual function. *Phys Rev* 85:845–81.
A review of the many functions of the retinal pigment epithelium and their role in various disease processes.
- Vila A, Satoh H, Rangel C et al 2012 Histamine receptors of cones and horizontal cells in Old World monkey retinas. *J Comp Neurol* 20:528–43.
- Wallman J, Winawer J 2004 Homeostasis of eye growth and the question of myopia. *Neuron* 43:447–68.
A review of the mechanisms of emmetropization and the causes of myopia.
- Wang D, Qi M, He M et al 2012 Ethnic differences of anterior chamber area and volume and its association with angle width. *Inv Ophthalmol Vis Sci* 53:3139–44.
An optical coherence tomography (OCT) investigation showing differences in the morphology of the anterior chamber between Caucasians and people with a Chinese heritage.
- Wässle H 2004 Parallel processing in the mammalian retina. *Nature Rev Neurosci* 5:747–57.
- Watson PG, Young RD 2004 Scleral structure: organisation and disease. A review. *Exp Eye Res* 278:609–23.
- Zhang HR 1994 Scanning electron-microscopic study of corrosion casts on retinal and choroidal angioarchitecture in man and animals. *Prog Retinal Eye Res* 13:243–70.